

---

# Parameter Estimation of Binned Hawkes Processes

Mathematical modelling of complex systems

---

*Written by*

Hana NAJI  
Bilel HATMI  
Tagne BONJAWO  
Titouan SIRE  
Tomas ESPAÑA

Students of CentraleSupélec (Université Paris-Saclay)

*Supervising professor*

Ioane MUNI TOKE

Permanent Faculty of the Chair of Quantitative Finance of CentraleSupélec

*Acknowledgements*

We are very grateful to Ioane Muni Toke for his careful help, comments and suggestions all along the development of this paper. All remaining errors are our own.

February 23, 2026

# Contents

|                                                                              |           |
|------------------------------------------------------------------------------|-----------|
| <b>Contents</b>                                                              | <b>1</b>  |
| Abstract . . . . .                                                           | 1         |
| Introduction . . . . .                                                       | 1         |
| <b>1 Estimation of Parameters in One-Exponential Hawkes Processes</b>        | <b>2</b>  |
| 1.1 The Whittle estimation . . . . .                                         | 2         |
| 1.2 The Uniform and Sequential Method . . . . .                              | 3         |
| 1.2.1 The Uniform Method . . . . .                                           | 3         |
| 1.2.2 The Sequential Method . . . . .                                        | 3         |
| 1.3 Comparing the three different <i>Unif</i> methods . . . . .              | 4         |
| 1.4 Comparing all methods . . . . .                                          | 7         |
| <b>2 Estimation of Parameters in Two-Exponential Hawkes Processes</b>        | <b>14</b> |
| 2.1 The Unif method . . . . .                                                | 14        |
| 2.1.1 Principle . . . . .                                                    | 14        |
| 2.1.2 Adaptation to a Two-Exponential Hawkes Process . . . . .               | 14        |
| 2.2 The Whittle estimation . . . . .                                         | 15        |
| 2.2.1 Adaptation to a Two-Exponential Hawkes Process . . . . .               | 15        |
| 2.2.1.1 Computation of the Fourier transforms of the kernel . . . . .        | 15        |
| 2.2.1.2 Computation of the mean intensity . . . . .                          | 15        |
| 2.3 Results . . . . .                                                        | 15        |
| 2.3.1 Validation of the method . . . . .                                     | 16        |
| 2.3.1.1 Principle . . . . .                                                  | 16        |
| 2.3.1.2 Results obtained with the Unif method . . . . .                      | 16        |
| 2.3.1.3 Results obtained with the Whittle method . . . . .                   | 19        |
| 2.3.2 Comparing Whittle and Unif in the two dimensional case . . . . .       | 25        |
| <b>3 Supervised Learning</b>                                                 | <b>27</b> |
| 3.1 Supervised Method . . . . .                                              | 27        |
| 3.1.1 Model Architecture . . . . .                                           | 27        |
| 3.1.2 Different Learning Method . . . . .                                    | 27        |
| 3.1.3 Data Generation . . . . .                                              | 28        |
| 3.1.4 Estimation of $\alpha$ and $\beta$ . . . . .                           | 28        |
| 3.1.5 Training and Evaluation . . . . .                                      | 29        |
| 3.1.6 Results . . . . .                                                      | 29        |
| 3.1.6.1 Influence of $\eta$ range . . . . .                                  | 29        |
| 3.1.6.2 Comparing all methods . . . . .                                      | 31        |
| 3.1.6.3 Improving the neural network performance: the LSTM network . . . . . | 33        |
| <b>4 Fitting Real Financial Data</b>                                         | <b>35</b> |
| 4.1 Results for the MLE method . . . . .                                     | 35        |
| 4.2 Results with MLE_random with degraded data . . . . .                     | 37        |
| 4.2.1 Whittle estimation . . . . .                                           | 38        |
| 4.2.1.1 Choice of a time window . . . . .                                    | 38        |
| 4.2.1.2 Choice of the bin size . . . . .                                     | 38        |

|          |                                                |           |
|----------|------------------------------------------------|-----------|
| 4.2.1.3  | Results . . . . .                              | 39        |
| 4.2.2    | Unif Method . . . . .                          | 40        |
| <b>5</b> | <b>Conclusion</b>                              | <b>42</b> |
|          | <b>Appendix : Speeding up the optimization</b> | <b>43</b> |
|          | <b>Bibliography</b>                            | <b>46</b> |

## Abstract

Estimating the parameters of a model is a key issue in quantitative finance. In this paper, we focus on Parameter Estimation of Binned Hawkes Processes proposed by Leigh Shlomovich, Edward A. K. Cohen, Niall Adams, and Lekha Patel in 2022 [6]. The objective of this project is to implement the estimation of a two-exponential Hawkes process and examine a supervised method for estimating the parameters of a one-exponential Hawkes process. We will conduct a comparative analysis of the estimation results obtained from the supervised method, the Maximum Likelihood Estimation (MLE) method, as well as the Whittle and Unif methods, which were discussed in previous S6 and S7 projects. Furthermore, we plan to evaluate the fitting performance of the two-exponential Hawkes process on real empirical data. All our code is gathered in the following link [Gitlab](#).

## Introduction

Hawkes processes are self-exciting processes that are used to model many phenomena in all sorts of scientific fields. The strength of this discrete model comes from the fact that the happening of an event is influenced by past events. This is why, originally, Hawkes processes were introduced to model the aftershocks of earthquakes: aftershocks are more likely to occur a short time after the earthquake than a long time after. Hawkes processes are also used in neuroscience, social media, statistical finance and even for predicting terrorist attacks [1]. Our goal is to apply the estimation methods we have previously studied for parameter estimation in a one-exponential Hawkes process to the context of a two-exponential Hawkes process. We aim to implement and compare these different approaches to estimate the parameters of the two-exponential Hawkes process accurately. Subsequently, we will delve into the exploration of a novel approach for parameter estimation in a one-exponential Hawkes process using supervised learning. Our motivation for investigating this Supervised Learning method stems from its advantageous characteristic mentioned in [3], as it is less affected by variations in binsize. Consequently, this method has the potential for improved performance compared to alternative approaches.

To what extent are Hawkes processes relevant when applied to financial data? The aim of the paper [4] is to discuss the difficulty in obtaining significant fits when testing the explanatory power of Hawkes processes. The author argues that this is due in part to discontinuities in trading activity, such as those that occur when markets open and close. The author suggests that working with data from FX markets, which operate continuously for longer periods, may be more successful in obtaining significant fits. The study found that Hawkes processes can be used to model a large number of events (around 15000 on average) with a high degree of accuracy.

Therefore, Hawkes processes seem to be very legitimate for modeling specific financial data. This paper [7] also suggests that. It discusses the relevance of trades-through in the market order model and describes how these can be well- modeled using Hawkes processes, and how this can help to explain periods of high liquidity consumption in order books. In sum, Hawkes processes are highly applicable for modeling financial data due to their self-exciting nature, which imparts significant modeling power. Therefore, to conclude our project, we will attempt to fit our two-exponential Hawkes process to real-world financial data. By doing so, we aim to further validate the efficacy and relevance of Hawkes processes in capturing the dynamics of financial systems.

# Chapter 1

## Estimation of Parameters in One-Exponential Hawkes Processes

In this section we start by quickly reminding the theoretical principles of the estimation methods we use to estimate binned Hawkes processes. Then, we will compare all these methods and try to deduce an optimal strategy that strikes a fair balance between accuracy of estimations and limitations in data precision.

### 1.1 The Whittle estimation

Whittle estimation is based on the minimization of the log-spectral likelihood of a process. It was introduced by Peter Whittle in 1951 [2]. Next, we introduce several mathematical notions to understand the foundations of this method.

**Definition 1.1.1** (Binned count series). *The count series with binsize  $\Delta$  associated to a point process  $N$  is the process  $(X_t)_{t \in \mathbb{R}} = (N(t\Delta, (t+1)\Delta))_{t \in \mathbb{R}}$ , generated by the count measure on intervals of size  $\Delta$ .*

**Theorem 1.1.1.** *Let  $N$  be a stationary Hawkes process on  $\mathbb{R}$ , with kernel  $h$  and  $(X_t)_{t \in \mathbb{R}} = (N(t\Delta, (t+1)\Delta))_{t \in \mathbb{R}}$  the associated count series. Then  $X_t$  has a spectral density function given by*

$$f_{X_t}(w) = m\Delta \text{sinc}^2\left(\frac{w}{2}\right) \left| 1 - \tilde{h}\left(\frac{w}{\Delta}\right) \right|^2 \quad (1.1)$$

**Remark 1.1.1.** *The symbol  $\tilde{\cdot}$  denotes the Fourier transform:*

$$\tilde{\phi}(w) = \int_{\mathbb{R}} e^{-iws} \phi(s) \, ds$$

**Corollary 1.1.1.** *Let  $N$  be a stationary Hawkes process on  $\mathbb{R}$  and  $(X_k)_{k \in \mathbb{Z}} = (N(k\Delta, (k+1)\Delta))_{k \in \mathbb{Z}}$  the associated count series. Then  $X_k$  has a spectral density function given by*

$$f_{X_k}(w) = \sum_{j \in \mathbb{Z}} f_{X_t}(w + 2k\pi)$$

where  $f_{X_t}$  is the function defined in the previous proposition.

**Remark 1.1.2.** *This corollary shows it is important to distinguish  $f_{X_k}$  when  $k \in \mathbb{Z}$  and  $f_{X_t}$  when  $t \in \mathbb{R}$ .*

**Theorem 1.1.2** (Equation of the estimator). *Let  $\theta$  be an unknown parameter vector. In our case, we will have  $\theta = (\mu, \alpha_1, \alpha_2, \beta_1, \beta_2)$  where  $\mu, \alpha$  and  $\beta$  are the three parameters of our model we want to estimate. The optimisation problem verified by  $\theta$  is the following for a stationary linear process  $(X_k)_{k \in \mathbb{Z}}$  with spectral density  $f_\theta$*

$$\boxed{\text{minimize } \mathcal{L}_n(\theta) \text{ s.t } L\theta > d} \quad (1.2)$$

where

$$\mathcal{L}_n(\theta) = \frac{1}{4\pi} \int_{-\pi}^{\pi} (\ln f_\theta(w) + \frac{I_n(w)}{f_\theta(w)}) dw$$

is the log-spectral likelihood of the process and

$$I_n(w) = \frac{1}{2\pi n} \left| \sum_{k=1}^n X_k e^{-ikw} \right|^2$$

is the periodogram of the partial realisation  $(X_k)_{1 \leq k \leq n}$

**Remark 1.1.3.** *We can prove that the estimator  $\widehat{\theta}_n$  is asymptotically normal and has a speed convergence in  $n^{-\frac{1}{2}}$ . This means that the mean square error  $\|\widehat{\theta}_n - \theta_0\|_2^2$  tends to a linear function of slope -1 when  $N \rightarrow +\infty$  in log-log scale. Note that  $\theta_0$  is the real set of parameters of the Hawkes process. We will illustrate this convergence speed with numeric simulations later on.*

## 1.2 The Uniform and Sequential Method

In this section we present the uniform and sequential method proposed by Leigh Shlomovich [6]. The main idea of these two methods is to estimate the parameters of a binned Hawkes process using the Expectation-Maximization algorithm by transforming the binned Hawkes process into a real timestamped-Hawkes process. We do that by sampling events using an alternative distribution  $q$ . When dealing with Hawkes processes, the challenge is to effectively generate samples of latent event times that align with the given bin counts. The ideal distribution for sampling from is  $p(\mathcal{T} | \mathbf{N}, \Theta^i)$ , which represents the missing event times given the bin counts and model parameters. To ensure optimal efficiency and meaningful weighting, it is crucial that the alternative distribution  $q(\mathcal{T} | \mathbf{N}, \Theta^i)$  used for sampling is as similar as possible to the true distribution of timestamps.

### 1.2.1 The Uniform Method

The first possibility is to uniformly redistribute the events across a bin in order to generate a set of event times. The advantage of this method is the gain of time to redistribute the event uniformly across a bin. However, the uniform distribution is not close enough to  $p(\mathcal{T} | \mathbf{N}, \Theta)$ . Our goal is to find an alternative distribution, which captures the inter-bin behavior of the process.

We create a theoretically deterministic approximation of the expectation by sequentially finding modes of the alternative distribution and using these as the “samples.” In particular, these “samples” are obtained by maximizing the joint truncated density over each bin. Maximizing the joint density yields consistent times for each bin, and repeating this process sequentially for all observed bins generates a continuous time version of the binned Hawkes process. In this way, this algorithm allows us to more easily use a distribution closer to the true distribution  $p(\mathbf{N}, \Theta)$ .

### 1.2.2 The Sequential Method

The sequential method aims to sample the timestamps in a more complex and closer manner to the dynamics of Hawkes processes. We suppose having already simulated until time  $t_n$ , with  $n \in 1, \dots, N(T)$  where  $N(T)$  is the total number of time events. In the following non empty bin, there are exactly  $m$  time-points, being  $t_{n+1}, \dots, t_{n+m}$ . The joint density of these events can be expressed as :

$$\begin{aligned} f_{T_{n+1}, \dots, T_{n+m}}^*(t_{n+1}, \dots, t_{n+m}) &= \prod_{i=1}^m f_{T_{n+i}}^*(t_{n+i}), \\ &= \prod_{i=1}^m \lambda^*(t_{n+i}) \exp\left(-\int_{t_{n+i-1}}^{t_{n+i}} \lambda^*(u) du\right) \end{aligned}$$

In our case of an exponential, kernel we can express this as :

$$\Sigma_{i=1}^m (\nu + \alpha A(n+i)) \exp(-\nu(t_{n+m} - t_n)) \exp(\Sigma_{i=1}^m \Sigma_{j=1}^{n+i+1} \frac{\alpha}{\beta} (e^{-\beta(t_{n+i}-t_j)} - e^{-\beta(t_{n+i-1}-t_j)}))$$

Where

$$A(n+i) = \Sigma_{j=1}^{n+i-1} \exp(-\beta(t_{n+i} - t_j))$$

We want to know the events times stamps  $t_{n+1}, \dots, t_{n+m}$  given the observed counts. Moreover, each time stamps is know to have occurred in our bin  $[b^-, b^+]$ . That's why we consider the truncated joint density. For this, we consider the cumulative distribution function (CDF) of the event times, and we require that  $b^- < t_{n+1} < \dots < t_{n+m} < b^+$ .

Therefore, we truncate the PDF by considering the joint CDF. As with the joint PDF, we can use factorization to express the joint CDF as

$$F_{T_{n+1}, \dots, T_{n+m}}^*(t_{n+1}, \dots, t_{n+m}) = \Pi_{i=1}^m F_{T_{n+i}}^*(t_{n+i})$$

where we can use the known form for the CDF of each successive time-point given the history of the process. That is, the joint CDF of time-points  $t_{n+1}, \dots, t_{n+m}$  is given by

$$F_{T_{n+i}}^*(t_{n+i}) = \Pi_{i=1}^m (1 - \exp(-\int_{t_{n+i-1}}^{t_{n+i}} \lambda^*(u) du))$$

In our case of an exponential kernel, the expression is :

$$\Pi_{i=1}^m (1 - \exp(-\int_{t_{n+i-1}}^{t_{n+i}} \nu + \Sigma_{j=1}^{n+i-1} \alpha \exp(-\beta(u - t_j)) du))$$

Thus, the joint truncated probability function of  $m$  time-points given the history  $t_1, \dots, t_n$  is expressed as :

$$\frac{f_{T_{n+1}, \dots, T_{n+m}}^*(t_{n+1}, \dots, t_{n+m})}{F_{T_{n+1}, \dots, T_{n+m}}^*(t_{n+1}, \dots, t_{n+m-1}, b^+) - F_{T_{n+1}, \dots, T_{n+m}}^*(b^-, t_{n+2}, \dots, t_{n+m})}$$

We assign the set of  $m$  proposed time stamps  $\tilde{t}_{n+1}, \dots, \tilde{t}_{n+m}$  for the given bin to be those that maximize the probability density function. In this way we can sequentially simulate a continuous version of the observed binned Hawkes process by progressively handling each bin such that we jointly maximize this likelihood.

### 1.3 Comparing the three different *Unif* methods

In this section, we give a quick reminder of the results found last semester with a few additional piece of information.

In our previous work, we focused on the following estimation methods:

1. **MLE**: Maximum Log-likelihood Estimation performed by passing the timestamps of a Hawkes process. It is our reference estimation method since it is the most precise.
2. **Unif\_paper**: estimation proposed in [6] based on a Monte-Carlo method and the Expectation-Maximization algorithm.
3. **Unif\_equal\_weights**: same as **Unif\_paper** except that all the weights given to the Hawkes processes generated during the Monte Carlo step are equal.
4. **Unif**: An update of our own of **Unif\_paper** which implements weights differently and seems more performant.
5. **Seq1**: sequential estimation proposed in [6] based on maximization of joint truncated density.
6. **Whittle**: The Whittle estimation, studied in one of our previous reports.

We recall that we specifically focus on binned Hawkes processes, meaning that if several events occur in a same bin, we do not know their distribution inside the bin, we only know how many events happened. Except **MLE**, which will be our reference method, all the other methods listed previously offer different solutions to estimate the parameters of such Hawkes processes. In this section, we focus on the set of three *Unif* methods.

The main idea of the *Unif* methods is to uniformly redistribute the timestamps across a bin and do as if the time series thus obtained is the «real initial» Hawkes process we want to estimate. More precisely, the idea is to generate several time series following this method (Monte-Carlo step of the algorithm) and give a weight to each one of them. The biggest weights are given to the time series that maximize the log-likelihood (acting as if they were Hawkes processes) with the current parameters. The new parameters are the ones that maximize the weighted mean of log-likelihoods of these generated time series. Here is how the algorithm **Unif\_paper** presented in [6] works:

---

**Algorithm 1** The **Unif\_paper** parameter estimation

---

**Require:**  $r, \mathcal{T}^*$   
 $\Theta^0 \leftarrow \text{sort}(\text{unif}(1,3))$   
 $i \leftarrow 0$   
**while** not converged **do**  
  **for**  $j = 1$  **to**  $r$  **do**  
     $\mathcal{T}^{(j)} \leftarrow$  uniformly generated time series  
     $w_j \leftarrow -\log\_lik(\mathcal{T}^{(j)}, \Theta^i) - \log\_dens$   
  **end for**  
  **for**  $j = 1$  **to**  $r$  **do**  
     $w_j \leftarrow \exp(w_j - \min_{i \in \{1, \dots, r\}} w_i)$   
     $w_j \leftarrow w_j / \sum_{k=1}^r \exp(w_k)$   
  **end for**  
   $\Theta^{i+1} \leftarrow \text{argmin}_{\Theta} -\frac{1}{r} \sum_{k=1}^r w_k \log\_lik(\mathcal{T}^{(k)}, \Theta^i)$   
   $i \leftarrow i + 1$   
**end while**  
**return**  $\Theta^i$

---

- The algorithm requires  $r$ , the number of Monte-Carlo loops and  $\mathcal{T}$  the binned Hawkes process.
- $\Theta$  is the estimate of the three parameters  $\mu, \alpha, \beta$ .  $\Theta^0$  is initialized randomly in  $[0, 1]^3$
- $\log\_lik(\mathcal{T}, \Theta)$  is the log-likelihood of time series  $\mathcal{T}$  as if it were a Hawkes process with parameters  $\Theta$ .
- $\log\_dens$  is the joint distribution of the binned Hawkes process.

Here are some remarks on how the weights are computed in **Unif\_paper** and how the method can be improved. First of all, the joint density is subtracted to all the weights but, the joint density is a constant so it doesn't seem relevant considering this quantity, especially since it computes very heavy factorials that Python can't support. Thus, this was removed in our proposition of parameter estimation **Unif**. Second, in the first step of the algorithm, the weights take the opposite value of the log-likelihood and then they are subtracted to the minimum weight in order to introduce an idea of distance between weights. However, we seek to maximize the weight of the Monte-Carlo simulation that obtained the highest log-likelihood. But, if we consider the opposite of the log-likelihood we are precisely giving the lowest weight to the simulation that obtained the highest log-likelihood. This doesn't seem correct. Thus, in the **Unif** method we propose, line 7 of the algorithm above becomes

$$w_j \leftarrow \log\_lik(\mathcal{T}^{(j)}, \Theta^i) \quad (1.3)$$

instead of

$$w_j \leftarrow -\log\_lik(\mathcal{T}^{(j)}, \Theta^i) - \log\_dens \quad (1.4)$$

Finally, we also introduced **Unif\_equal\_weights** where all the weights are equal. The idea was to measure until what extent the weights were relevant in this method. Our choice of weights was

$$w_j \leftarrow \frac{1}{r} \quad (1.5)$$

Now, let's see how **Unif**, **Unif\_paper** and **Unif\_equal\_weights** perform (for more details, see our previous report). The following graphs are boxplots of 100 simulations each for parameters  $\mu, \alpha, \beta$  and  $\Delta = 1$  (binsize) when the time horizon of the Hawkes process increases.

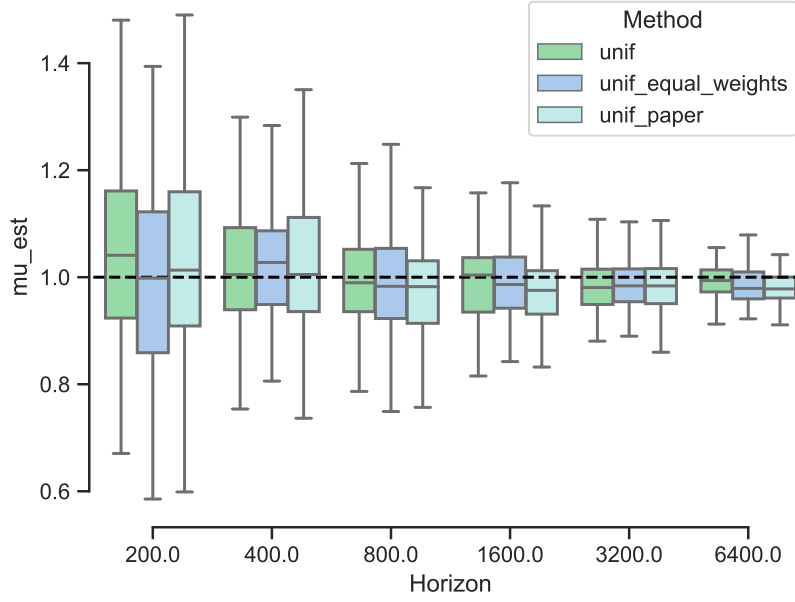


Figure 1.1: **Estimation of  $\mu$**

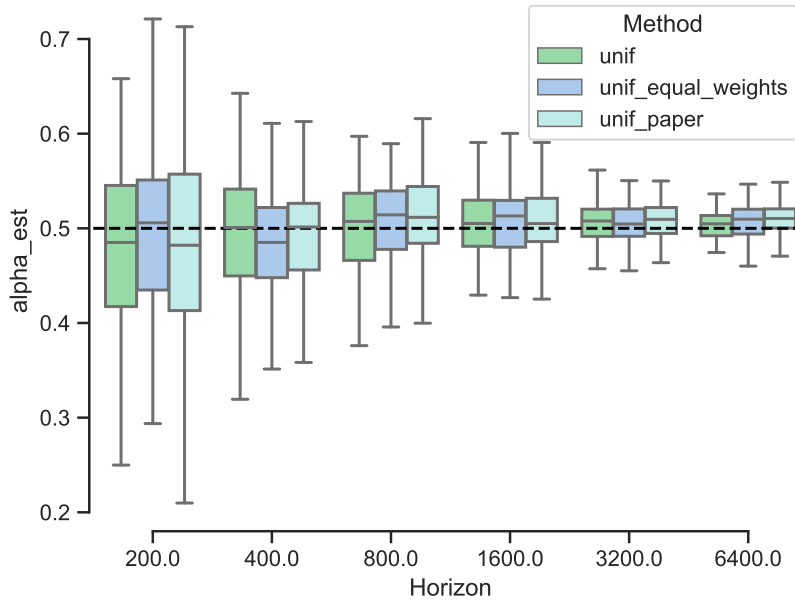


Figure 1.2: **Estimation of  $\alpha$**

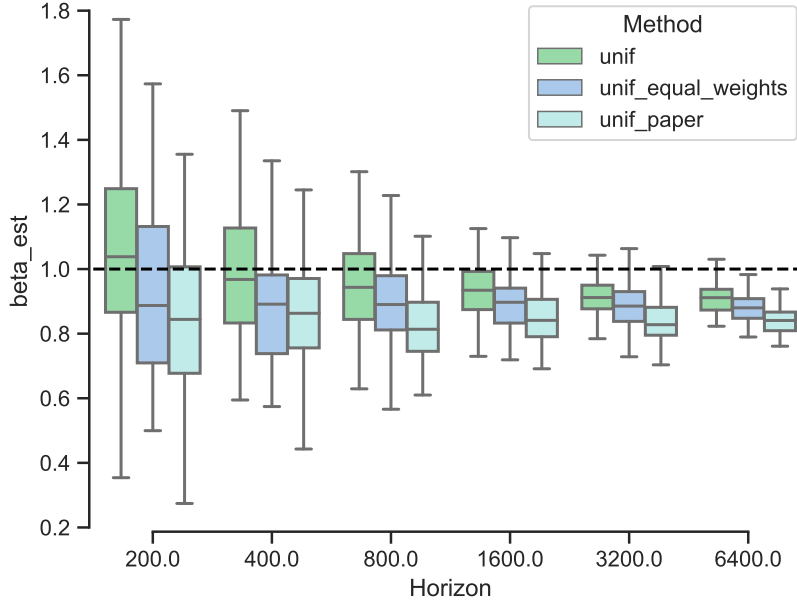


Figure 1.3: **Estimation of  $\beta$**

Here are the important remarks we can make:

- The boxplots get thinner as the time horizon increases. This is expected since the more points we have, the more precise the estimation gets.
- For  $\mu$  and  $\alpha$ , all three methods perform well but systematically **Unif** performs better with slightly less bias.
- For  $\beta$ , the performance of all three methods degrades very quickly. None of the three can be used to estimate this parameter, especially as the time horizon increases. However, we note that **Unif** performs systematically better than the two others.
- Giving equal values to all weights performs very well for  $\alpha$  and  $\beta$ . One can intuit that this comes from the size of the bin: here  $\Delta = 1$  which is very small compared to the time horizon. Thus, uniformly distributing events in this small bin won't have a significant impact on the log-likelihood. Surprisingly, it has a much bigger impact for the estimation of  $\beta$  where **Unif** performs better than **Unif\_equal\_weights** which itself also performs better than **Unif\_paper**. But anyways, the estimations are biased so the results can't be exploited.

To summarize, these results seem to justify and validate the changes we have made to come up with **Unif**. For this reason, we will only consider this method in what follows.

## 1.4 Comparing all methods

Here we show the boxplots in order to compare all the estimation methods we studied. The parameters take the following values:  $\mu = 1.0$ ,  $\alpha = 0.5$  and  $\beta = 1.0$ . Note that this time, the time horizon is fixed to a large value ( $T = 3200$ ) and the bin size varies. The **MLE** estimation is made with the exact timestamps of the simulated Hawkes processes and serves as a reference to compare all the other methods that estimate binned Hawkes processes.

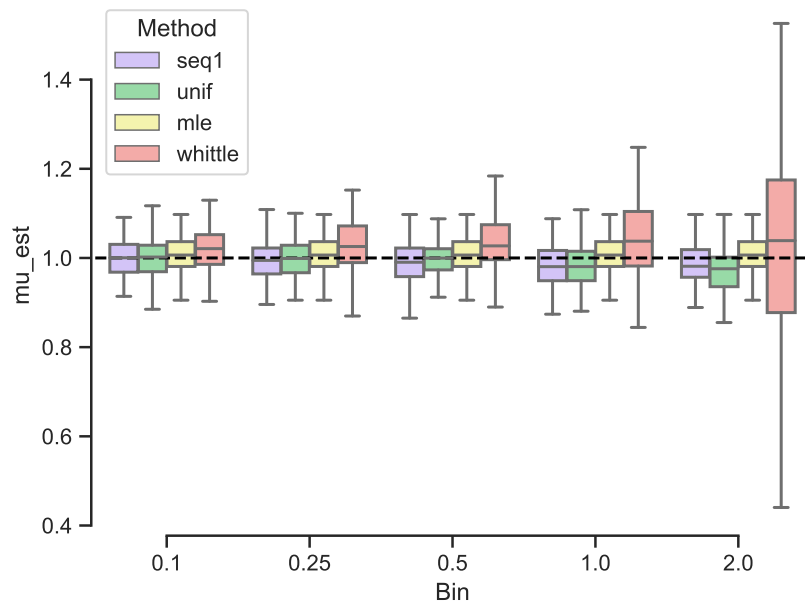


Figure 1.4: Estimation of  $\mu$

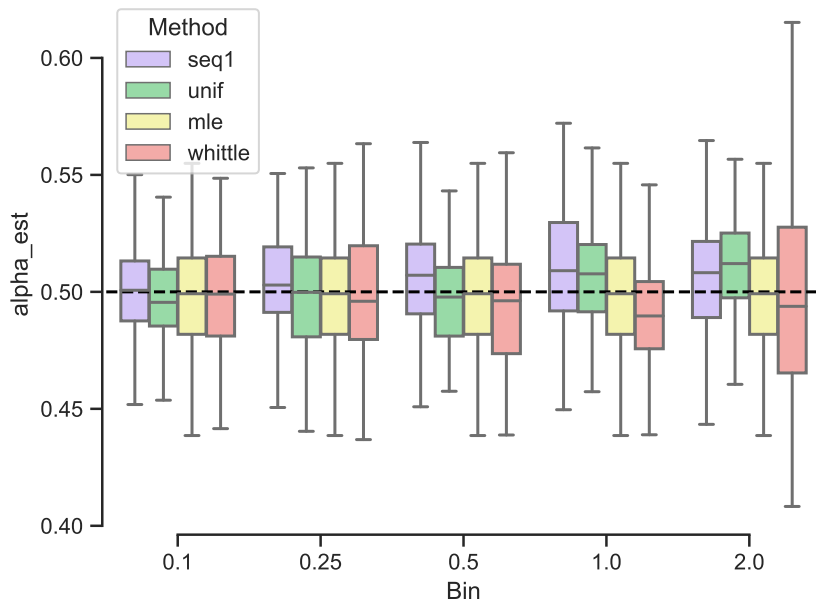


Figure 1.5: Estimation of  $\alpha$

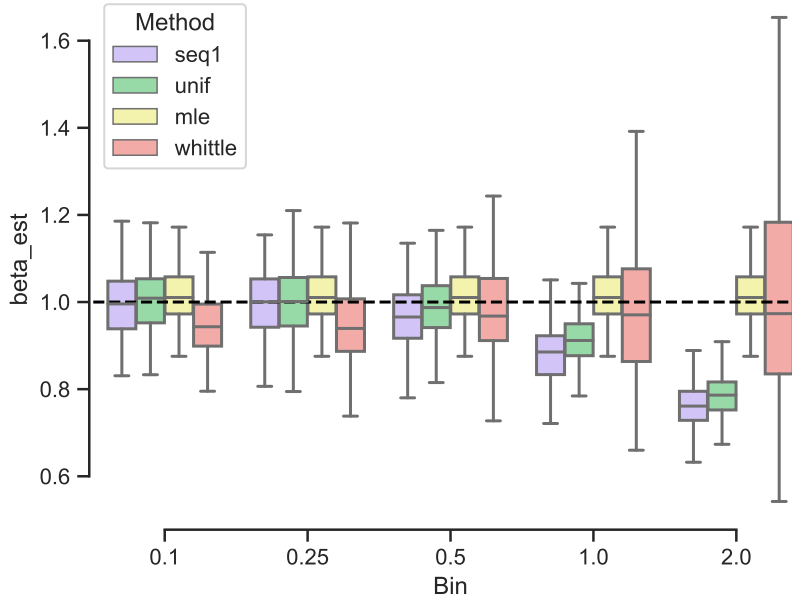


Figure 1.6: **Estimation of  $\beta$**

Here are some remarks we can make:

- For  $\mu$ , the bin doesn't really degrade the quality of the estimation, except for the Whittle estimation which has a bias in addition to degraded precision. **Unif** and **Seq1** have similar performances than **MLE**, except for a slight bias that appears for a bin size bigger than one. Still this remains very satisfactory and **Unif** and **Seq1** perform well to estimate  $\mu$  for a bin size between 0.1 and 2.0.
- For  $\alpha$ , the results are pretty similar except for Whittle estimation that considerably gains in precision. The results are very satisfactory for the three methods for bins smaller than 0.5. More generally, we can even consider all three methods acceptable for bins ranging from 0.1 to 2.0.
- For  $\beta$ , the results are very satisfactory for **Unif** and **Seq1** when the bin is smaller than 0.5 and strongly biased when it is bigger than 1.0. Only the Whittle estimation is reliable in the case of bigger bins: the median is close to the correct value but the variance is important. We note that surprisingly, the bias of the Whittle estimation tends to decrease as the bin increases which is the total opposite of the two other methods. We believe that this is due to noise in the data and shouldn't be considered as a general property.

To summarize, the choice of the method really much depends on how precise one wants the estimates to be and also of the power of computations available. For parameters  $\mu$  and  $\alpha$ , one can achieve very precise estimations that are as accurate as **MLE** for all three methods for small bins. For parameter  $\beta$ , estimations are also as accurate as **MLE** but not for the **Whittle** method anymore and only for small bins. For bigger bins **Unif** and **Seq1** become strongly biased and can't be used. In this case, the **Whittle** method can be used but the variance will be important. Regarding the available power of computations, it is also very important to take into account the time execution of each method. While **MLE** is performed almost instantaneously, the other methods are very long to execute, especially **Seq1**. Unfortunately, we didn't gather precise time executions but, to give an order of magnitude, the estimations had to be performed on the supercomputer of the university because our computers couldn't support such calculations. On the contrary, all estimations with **MLE** were almost instantaneous and could be performed on our computers.

In this paragraph we will generalize the notion of "small" and "big" bin we mentioned previously. Indeed, we noticed that the estimations were very accurate for "small" bins, i.e. smaller

than 0.5. But the notion of "small" bin directly depends on the parameters of the Hawkes process. Indeed, it is not the size itself of the bin that is important but the number of events in each bin. Intuitively, the more events the bins have, the more the estimation methods will struggle to "capture" the information of the Hawkes process and the badder the estimates. Thus, the most relevant piece of information is the mean number of events in non empty bins. Since we focus on binned Hawkes processes our goal is to strike a fair balance between precision of estimates and number of events per non empty bins. In the following graphs, we show the histogram of events per non empty bins for  $\mu = 1.0, \alpha = 0.5, \beta = 1.0, T = 3200$  as the bin size changes.

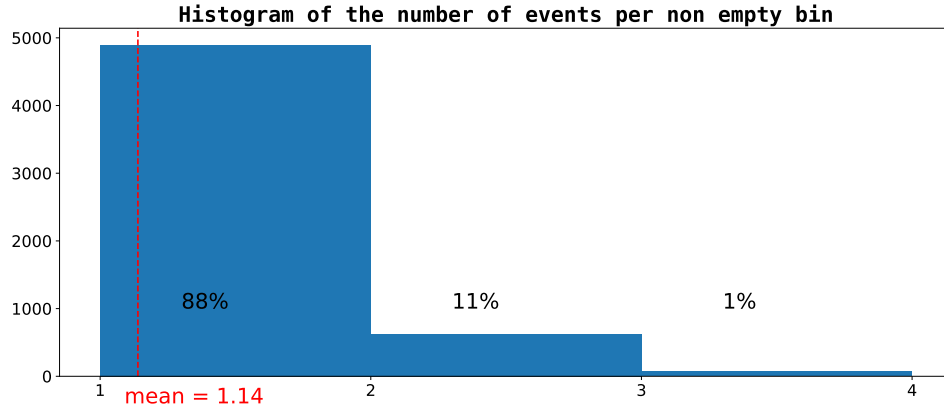


Figure 1.7:  $\Delta = 0.1$

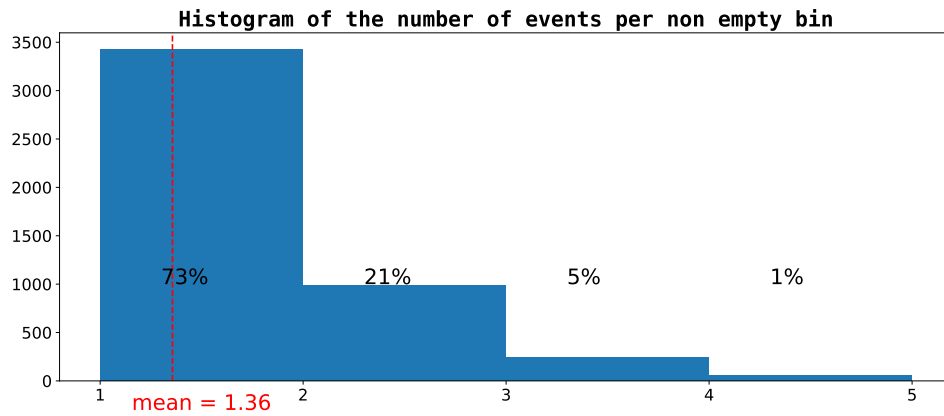


Figure 1.8:  $\Delta = 0.25$

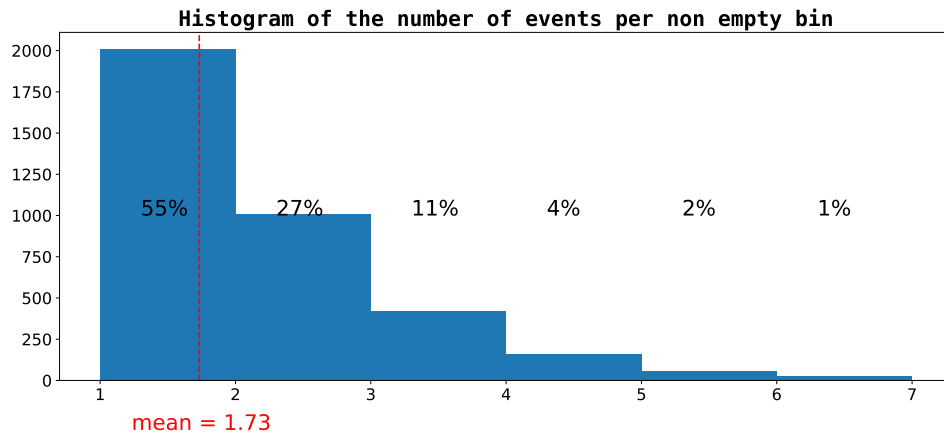


Figure 1.9:  $\Delta = 0.5$

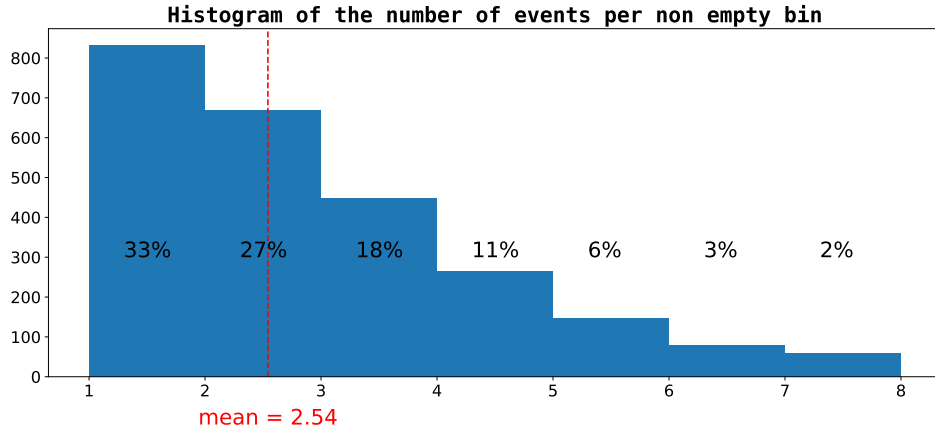


Figure 1.10:  $\Delta = 1.0$

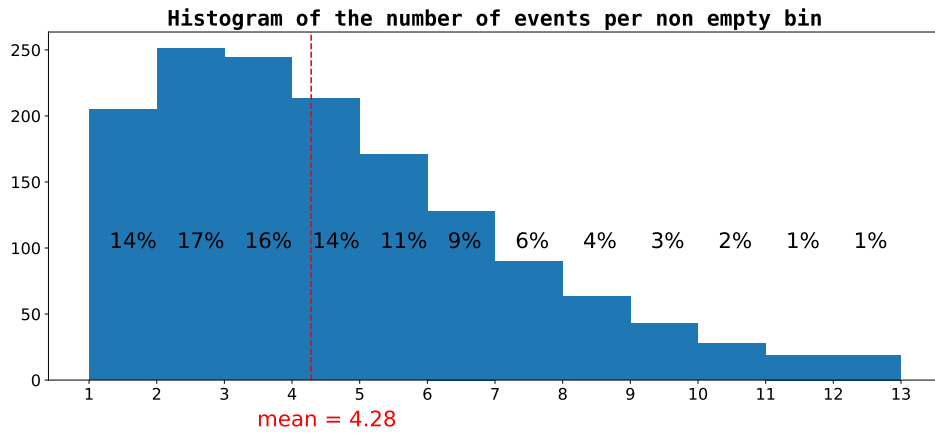


Figure 1.11:  $\Delta = 2.0$

These graphs help us to understand why  $\Delta = 0.5$  was a threshold to have accurate precisions. Indeed, we note that  $\Delta = 1.0$  is the first bin size for which the mean of events per non empty bin becomes bigger than two. This means that our estimation methods start loosing precision when there are more than two events per non empty bin in terms of mean. This of course is intuitive but it also questions the usefulness of the estimation methods we study: what is the point of studying methods that specifically apply to binned Hawkes processes if there precision starts to strongly degradate as soon as the mean number of events per non empty bin becomes greater than two (especially when we know that their time complexity is much bigger than **MLE**)? To try to answer this question we will focus on a new method. We will call it **MLE\_random** standing for MLE randomized. This method simply redistributes uniformly the events in the bin and then applies **MLE** to the time series thus obtained. Lets see how it performs.

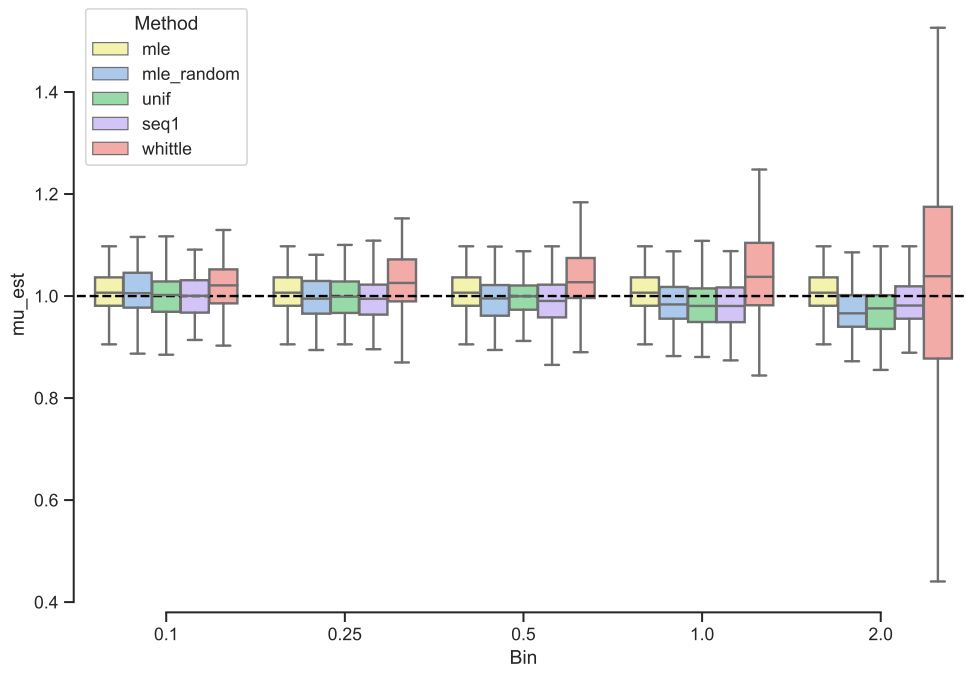


Figure 1.12: Estimation of  $\mu$

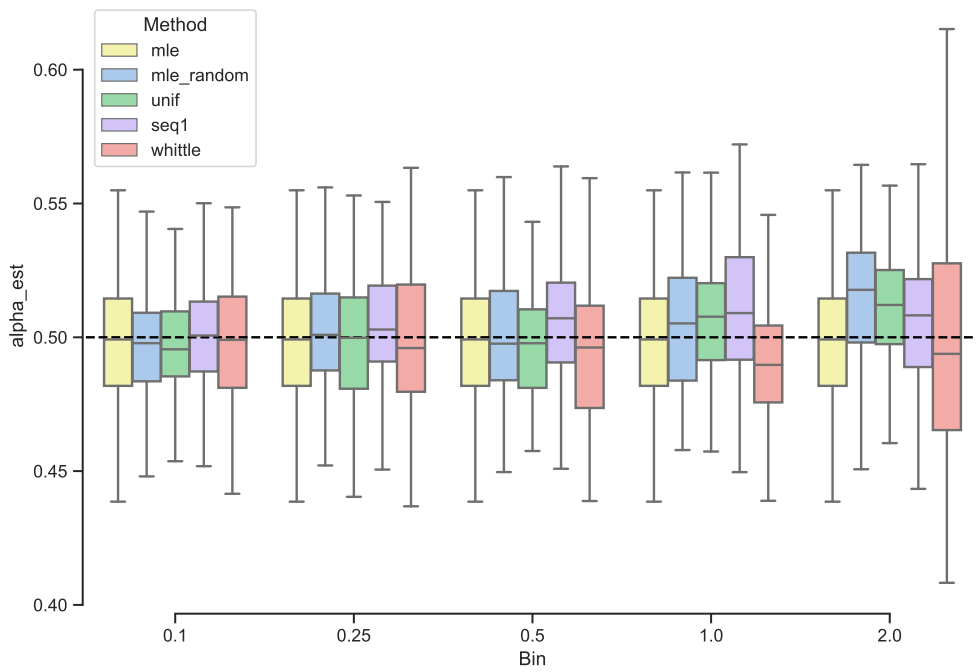


Figure 1.13: Estimation of  $\alpha$

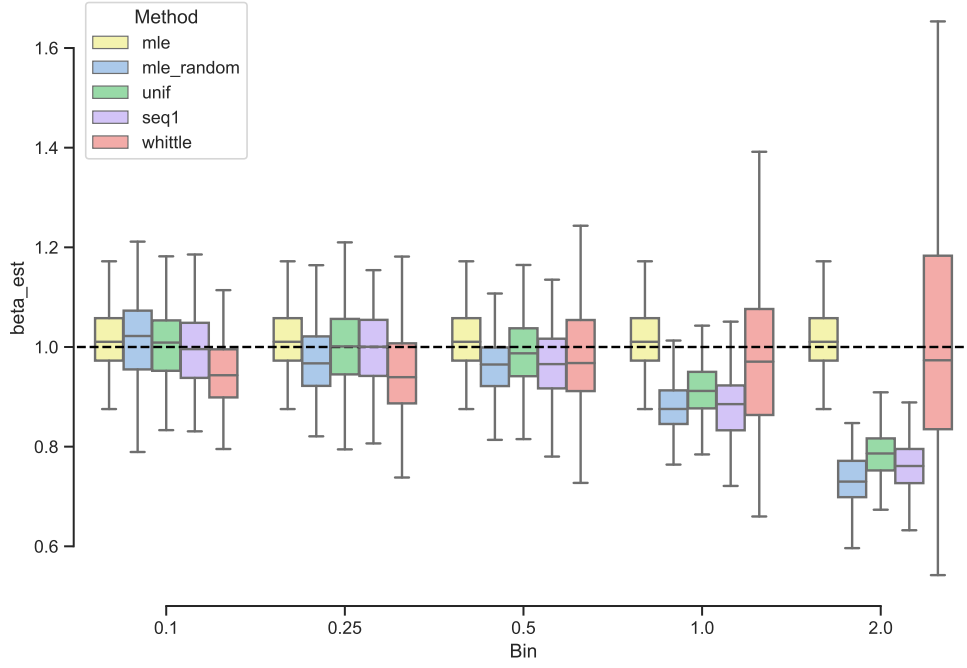


Figure 1.14: **Estimation of  $\beta$**

We remark that **MLE\_random** has the same degree of performance than both **Unif** and **Seq1**. But, its computations are much faster which clearly makes it the best method to estimate the parameters of binned Hawkes processes if one can afford to have at most 2 events per non empty bin in mean. If this is not the case, then the optimal strategy could consist in estimating  $\mu$  and  $\alpha$  with **MLE\_random** (precise and quick execution) and  $\beta$  with **Whittle** (unique method that isn't biased for more than two events per non empty bin). Further estimations not shown here show that the **Whittle** estimation becomes strongly biased for more than 5 events in mean per non empty bin, showing that there is an upper bound to be respected if we want to keep an acceptable bias.

## Chapter 2

# Estimation of Parameters in Two-Exponential Hawkes Processes

The reason for using a two-exponential Hawkes process is to capture the temporal clustering or burstiness observed in real-world event sequences. In many real-world systems, events tend to occur in clusters or bursts rather than being uniformly distributed over time. This bursty behavior is often characterized by two distinct time scales: a short time scale for immediate clustering of events following an occurrence, and a longer time scale for the decay of clustering effects.

By incorporating two exponential decay terms in the Hawkes process, the two-exponential model allows for a more flexible representation of the clustering behavior. The first exponential term captures the short-term clustering effect immediately following an event, while the second exponential term captures the longer-term decay of clustering effects. This enables the model to capture both the immediate impact and the lasting influence of previous events on the occurrence of future events.

Thus, the intensity  $\lambda$  of a two-exponential Hawkes process can be written as :

$$\lambda(t) = \mu + \sum_{l=1}^2 \sum_{t_i < t} \alpha_l \beta_l e^{-\beta_l(t-t_i)} \quad \text{such that } \mu \in \mathbb{R}_+, 0 < \sum_{l=1}^2 \alpha_l < 1 \text{ and } \beta_l \in \mathbb{R}_+.$$

## 2.1 The Unif method

### 2.1.1 Principle

The principle of the estimation of the parameters of a two-exponential hawkes process remains the same as to the estimation of the parameters of a classic one-exponential hawkes process. As said earlier in this report, it consists in uniformly redistributing the timestamps across a bin and do as if the time series thus obtained is the «real initial» Hawkes process we want to estimate. The parameters are the ones that maximize the weighted mean of log-likelihoods of these generated time series. The difference of the estimation of the parameters in between the two-exponential and the single exponential hawkes processes lies in the way of calculating the log-likelihood.

### 2.1.2 Adaptation to a Two-Exponential Hawkes Process

The expression of the log-likelihood is given by

$$l = \sum_{i=1}^k \ln(\lambda(t_i)) - \int_0^T \lambda(s) ds$$

where the  $t_i$  are the timestamps of the Hawkes process of conditional intensity  $\lambda$  given by

$$\lambda(t) = \mu + \sum_{l=1}^2 \sum_{t_i < t} \alpha_l \beta_l e^{-\beta_l(t-t_i)}$$

As the two-exponential Hawkes process has  $(\mu, \alpha_1, \alpha_2, \beta_1, \beta_2)$  as parameters, we can then transform the expression above to obtain :

$$l(\mu, \alpha_1, \alpha_2, \beta_1, \beta_2) = \sum_{i=1}^k \ln(\mu + \sum_{l=1}^2 \alpha_l \beta_l \sum_{j=1}^{i-1} e^{-\beta_l(t_i - t_j)}) - \mu T + \sum_{l=1}^2 \alpha_l \sum_{i=1}^k (e^{-\beta_l(T - t_i)} - 1)$$

By replacing  $\sum_{j=1}^{i-1} e^{-\beta_l(t_i - t_j)}$  with  $A_l(i)$  and by setting  $A_l(1) = 0$  the latter expression can be written more elegantly as:

$$l(\mu, \alpha_1, \alpha_2, \beta_1, \beta_2) = \sum_{i=1}^k \ln(\mu + \sum_{l=1}^2 \alpha_l \beta_l A_l(i) - \mu T + \sum_{l=1}^2 \alpha_l \sum_{i=1}^k (e^{-\beta_l(T - t_i)} - 1))$$

Where the  $(A_l)_{l \in \{1,2\}}$  sequences are computed recursively exactly as in the one-dimensional case as :

$$A_l(i) = e^{-\beta_l(t_i - t_{i-1})}(1 + A_l(i-1))$$

(see the section 2.3 of our Whittle report for further details).

## 2.2 The Whittle estimation

### 2.2.1 Adaptation to a Two-Exponential Hawkes Process

#### 2.2.1.1 Computation of the Fourier transforms of the kernel

The kernel for a Two-Exponential Hawkes Process is

$$h(t) = \sum_{i=1}^2 \alpha_i \beta_i \exp(-\beta_i t) I_{t>0}(t)$$

Therefore we can compute the Fourier transform of the kernel and find

$$\tilde{h}(x) = \sum_{i=1}^2 \alpha_i \beta_i \frac{\beta_i - jx}{x^2 + \beta_i^2} \quad (2.1)$$

#### 2.2.1.2 Computation of the mean intensity

The remaining unknown is  $m$ , the mean intensity  $E(N(0, 1))$  for a Two-Exponential Hawkes Process.

$$m = \frac{\mu}{1 - \int_{\mathbb{R}} h(t) dt}$$

Therefore, the mean number of events per bins is

$$m = \frac{\mu}{1 - \sum_{i=1}^2 \alpha_i} \quad (2.2)$$

## 2.3 Results

In the estimation of parameters for the two-exponential Hawkes process, the initial conditions of the estimated parameters can have an impact on the accuracy and reliability of the estimation. These initial conditions can influence the convergence of the estimation method and may introduce bias in the estimated parameters.

To mitigate the potential bias arising from the choice of initial conditions we have decided to verify the estimation method by initializing the parameters with the real values of a simulated Hawkes process. This approach ensures that the estimated parameters are not influenced by initial conditions.

In the second part of this research, we will focus on analyzing the impact of different initial conditions on the estimation results. We will compare the performance of the estimation method with different initializations and we will carefully assess the influence of these different initial conditions on the estimated parameters and their implications for the reliability and accuracy of the estimation results.

## 2.3.1 Validation of the method

### 2.3.1.1 Principle

In our study, for on parameters setting, we conducted 50 simulations of the Hawkes process with estimated parameters, initializing the parameters with the real values of the simulated Hawkes process, specifically setting  $\mu = 1$ ,  $\alpha_1 = 0.2$ ,  $\alpha_2 = 0.4$ ,  $\beta_1 = 1$ , and  $\beta_2 = 10$ .

We are aware that the choice of bin width, or the discretization of the simulated Hawkes process, can have a significant influence on the estimation results. Therefore, we have also considered the impact of different bin widths on the estimation performance in our study. The bin widths we have considered are 0.1, 0.3, 0.5, 0.7, 1, and 2, which correspond to different "densities" or numbers of points in each non-empty bin.

By varying the bin widths, we aim to understand the influence of the binning process on the estimation of parameters in the two-exponential Hawkes process and assess the impact of bin width on potential biases. The analysis of different bin widths will allow us to gain insights into the optimal binning strategy for estimating parameters in the two-exponential Hawkes process and provide recommendations for selecting an appropriate bin width in practice.

We have generated box plots to visually represent the estimation results for each parameter in the two-exponential Hawkes process, using different bin widths. Each box plot provides a summary of the distribution of the estimated parameter values. In addition to the box plots, we have included a black line on each graph to represent the real value of the corresponding parameter in the simulated Hawkes process. This allows for a direct comparison between the estimated parameter values and the ground truth from the simulated process.

### 2.3.1.2 Results obtained with the Unif method

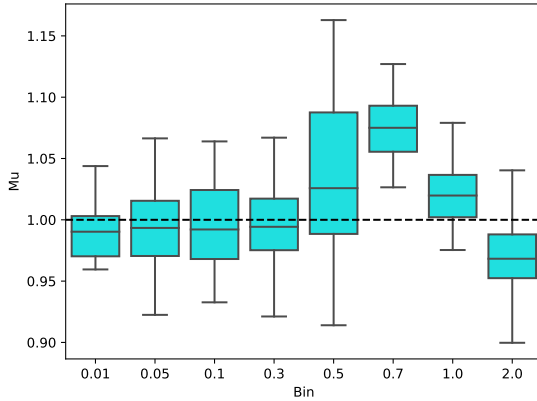


Figure 2.1: The estimation of  $\mu$  for different bin widths

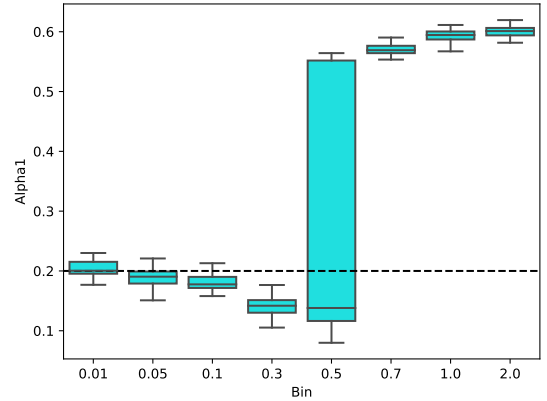


Figure 2.2: The estimation of  $\alpha_1$  for different bin widths

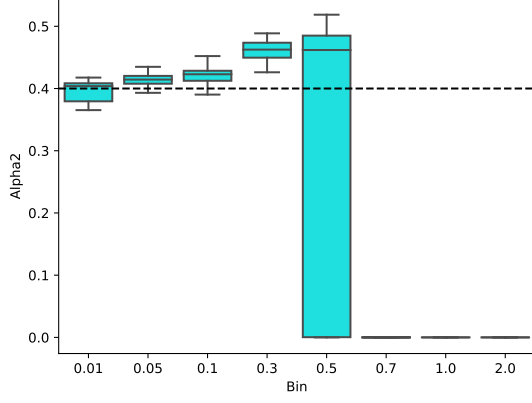


Figure 2.3: The estimation of  $\alpha_2$  for different bin widths

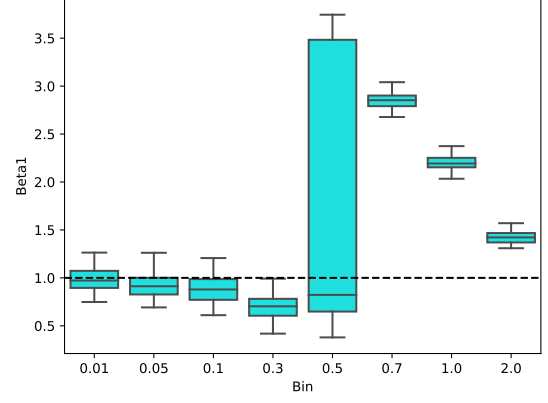


Figure 2.4: The estimation of  $\beta_1$  for different bin widths

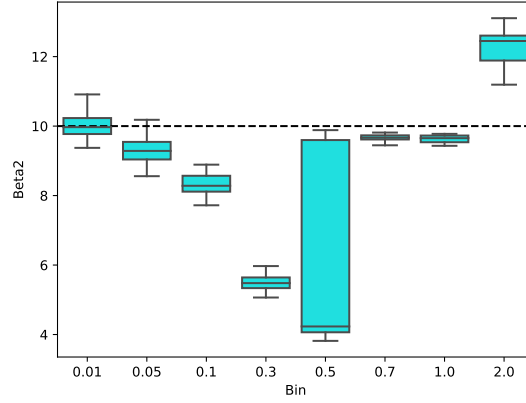


Figure 2.5: The estimation of  $\beta_2$  for different bin widths

Based on the figures and captions you provided, it seems that the bin width has a significant impact on the estimation of certain parameters, such as  $\alpha_1$  and  $\beta_1$ . When the bin width is larger than 0.5, the estimated values for  $\alpha_1$  and  $\beta_1$  become nonsensical, indicating that the choice of bin width is crucial for obtaining meaningful results.

Furthermore, there appears to be a noticeable difference in the estimated values when using different bin widths, such as 0.3 or 0.1. This suggests that the choice of bin width is important for the overall functionality of the estimation method. Specifically, it seems that the method is only functional for small bin widths, which implies a certain "density" of data points.

This observation highlights the significance of carefully selecting an appropriate bin width when using this estimation method. A bin width that is too large can result in inaccurate and nonsensical estimations, while a smaller bin width may yield more reliable results. It is crucial to consider the underlying data and the characteristics of the estimation method when determining an appropriate bin width for a particular analysis or application.

It may be worthwhile to further investigate the impact of bin width on the estimation results and assess the sensitivity of the method to different bin widths. This understanding can lead to more robust and accurate estimations in future analyses or applications that utilize this method.

**Remark 2.3.1.** Upon closer examination of the boxplots, it became evident that the two-dimensional nature of the process is not adequately captured when the bin size exceeds 0.6. In particular, Figure 2.6 clearly demonstrates that the alpha parameter is set to 0 after reaching the bin value of 0.6. Consequently, the process is treated as a one-exponential Hawkes process. Furthermore, when the number of events in a single bin surpasses 2 (specifically, bin 0.45 in this case), the estimations deteriorate significantly, resulting in an incomplete understanding of the complexity inherent in the two-exponential Hawkes process.

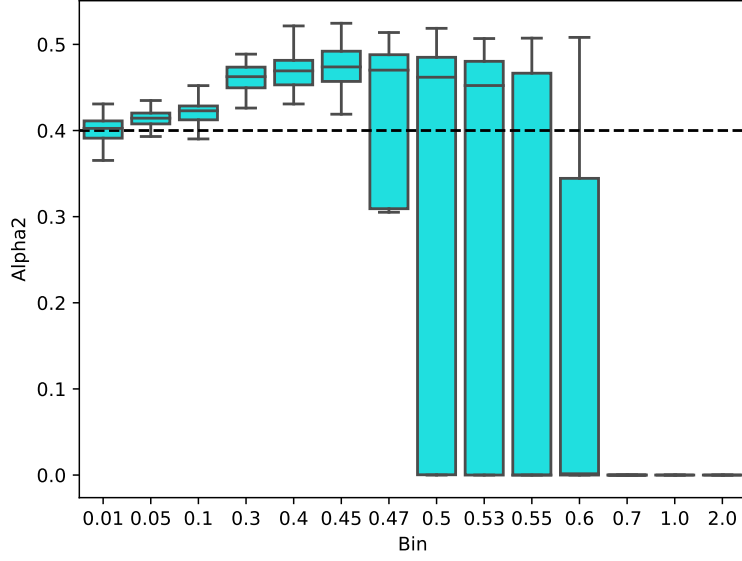


Figure 2.6:  $\alpha_2$  with more precision of the bin 0.5

#### Sensitivity of the estimate to initial conditions

The estimation method appears to work effectively for certain parameters (for  $bin = 0.1$ ). We examine if the algorithm is sensitive to initial conditions and if there are any potential limitations associated with them. In this analysis, we will investigate the impact of the initial conditions of  $\alpha_1$  and  $\alpha_2$  on the system's behavior. Our focus in this section will be primarily on  $\beta_1$  and  $\beta_2$ , while acknowledging that similar patterns and trends are observed for the other parameters.

We systematically selected six values for  $\alpha_1$  ranging from 0.1 to 0.3, and similarly, we chose six values for  $\alpha_2$  ranging from 0.2 to 0.6. It is worth noting that the true values for  $\alpha_1$  and  $\alpha_2$  are 0.2 and 0.4, respectively

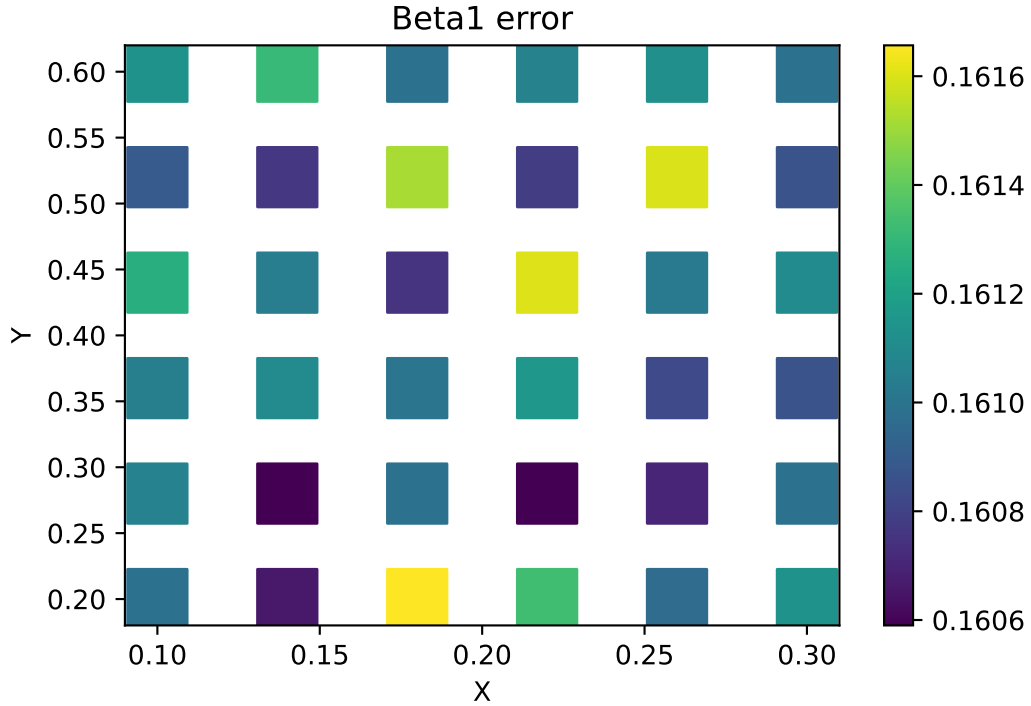


Figure 2.7: The impact of initial condition of  $\alpha_1$  and  $\alpha_2$  on the estimation of  $\beta_1$

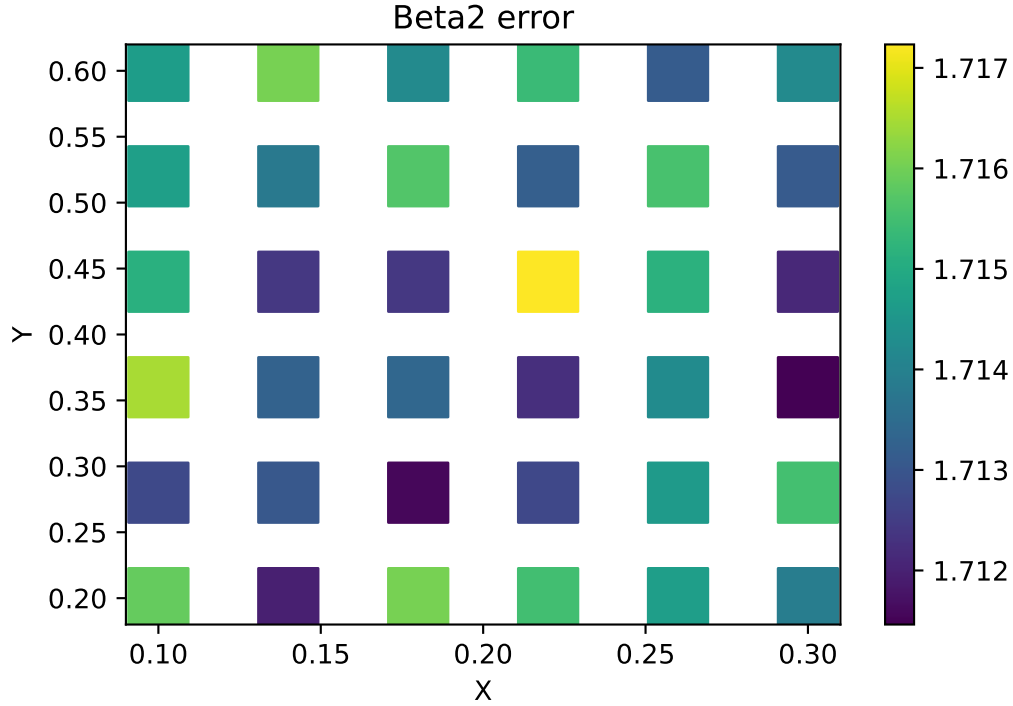


Figure 2.8: The impact of initial condition of  $\alpha_1$  and  $\alpha_2$  on the estimation of  $\beta_2$

In order to assess the impact of the degradations, we utilized the square error metric, which measures the squared difference between the exact parameters and the parameters obtained when initially perturbed. Figure 1 illustrates that the range of error values is 0.001, with a mean of 0.1610. This discrepancy is relatively small when compared to the value we are attempting to estimate, which is 1. Similarly, the error in estimating  $\beta_2$  amounts to a 17% difference relative to the parameter we seek to estimate, which is 10 in our specific case and with a range of error of 0.005. The small range of errors observed indicates that the estimation is not highly sensitive to the perturbation of the initial conditions we introduced.

### 2.3.1.3 Results obtained with the Whittle method

As it is shown in formula (3.4),  $0 < \sum_{i=1}^2 \alpha_i < 1$ . Therefore the estimation of the parameters is a constrained optimization problem. This is the reason why we used sequential linear-quadratic programming (SLQP) through the 'SLQP' method of the scipy.optimize library in the optimization of the whittle method. In fact, we first tried to estimate the parameters using the 'L-BFGS-B' of the library but estimates were often the bounds of the domain of definition. Then we tried to use genetic algorithms. But even with high value of cross-over, and mutation, we were often trapped by local minimum. Finally, we used the 'L-BFGS-B' method with a huge penalization to take into account the constraints because we had too much problems when the initial conditions were too far from the true values.

#### Starting from the exact initial conditions

The first thing we should verify is that when our initial conditions are equal very close to the true values, the algorithm converge to the true values of the process. That's why we simulated several hawkes processes (around 300), with a two-exponential kernel and we varied the size of the binsize among 0.02,0.05,0.1,0.2. Then we estimate the parameters using Witthle estimation and 'SLQP' optimization as the algorithm of constrained minimization.

The results are illustrated by the following boxplots:

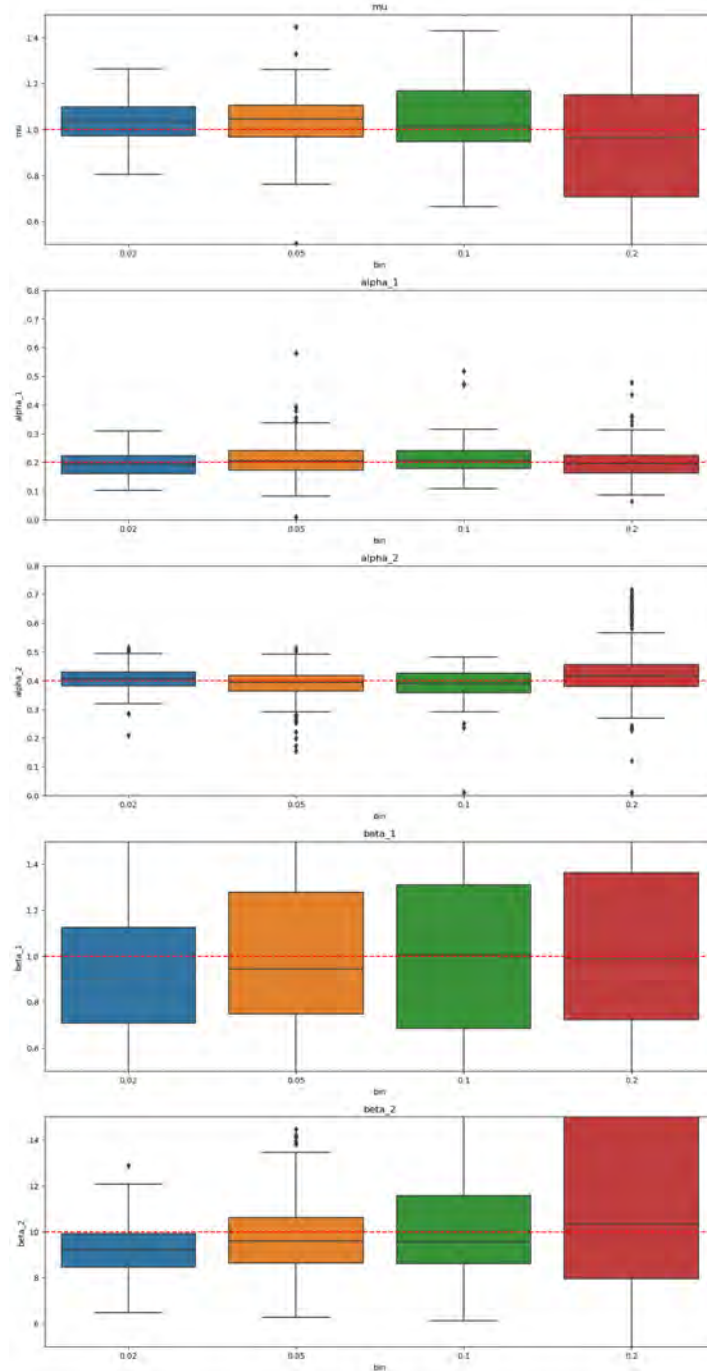


Figure 2.9: Sensitivity of the parameters estimate with perfect initial conditions

$\beta_2$  remains the parameter the most difficult to estimate with the greatest variance and bias. However, the relative error of the median estimator remains less than 10 % which is satisfying. We can see a degradation of the variance with the increase of the binsize. Furthermore, to minimize the bias of our estimator, it seems that a minimal binsize (here 0.02), is not the ideal choice. Indeed a binsize of 0.05 and 0.1 gives better results for our estimator. All in all, going very close from the true values allow satisfying estimation of each parameters.

### Sensitivity of the estimate to binsizes and limit size of the bin size

We already know that estimates deteriorate as binsize increases. We were therefore interested in the evolution of these degradations, and in the limit size we could take for the bin to keep a reasonable bias and variance. To do this, we estimated two Hawkes processes with respective parameters  $\mu = 1, \alpha_1 = 0.25, \alpha_2 = 0.25, \beta_1 = 1, \beta_2 = 10$  and  $\mu = 1, \alpha_1 = 0.25, \alpha_2 = 0.25, \beta_1 = 0.1, \beta_2 = 1$  and binsizes ranging from 0.02 to 10.

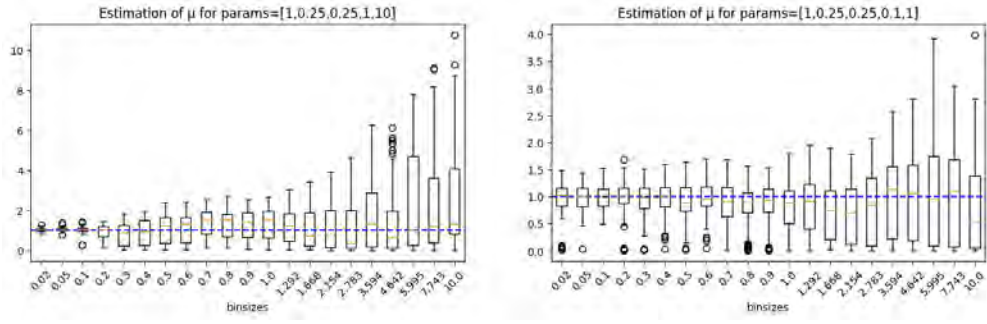


Figure 2.10: Sensitivity of  $\mu$  to binsizes

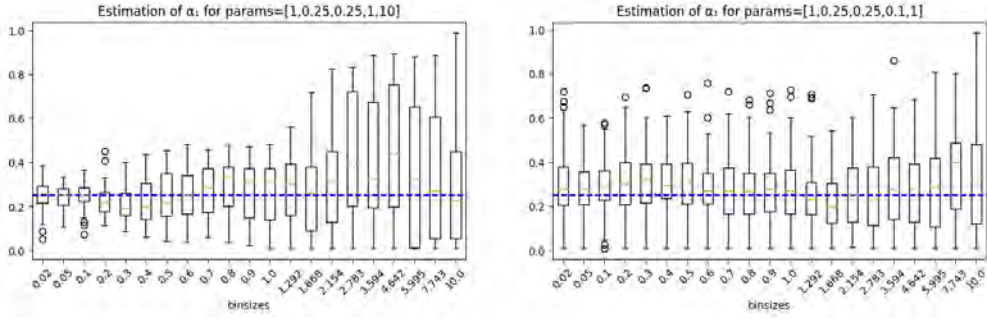


Figure 2.11: Sensitivity of  $\alpha_1$  to binsizes

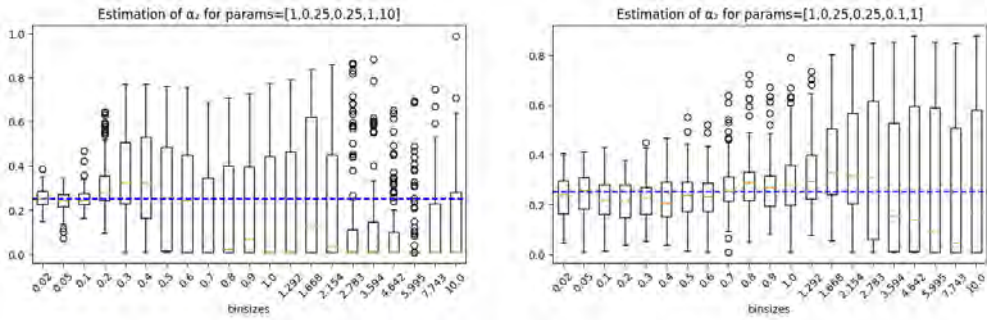


Figure 2.12: Sensitivity of  $\alpha_2$  to binsizes

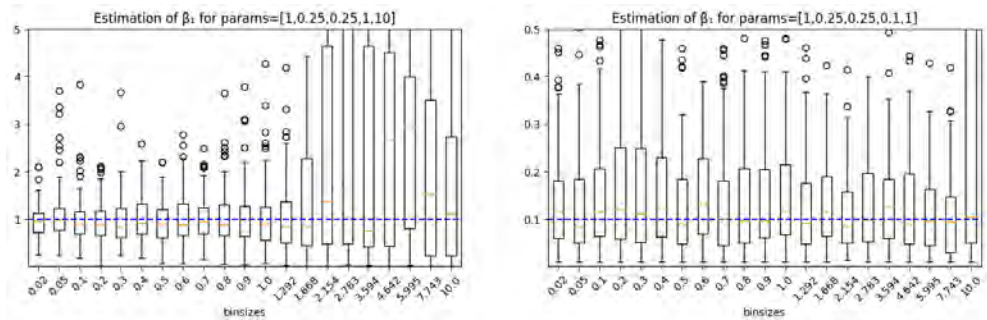


Figure 2.13: Sensitivity of  $\beta_1$  to binsizes

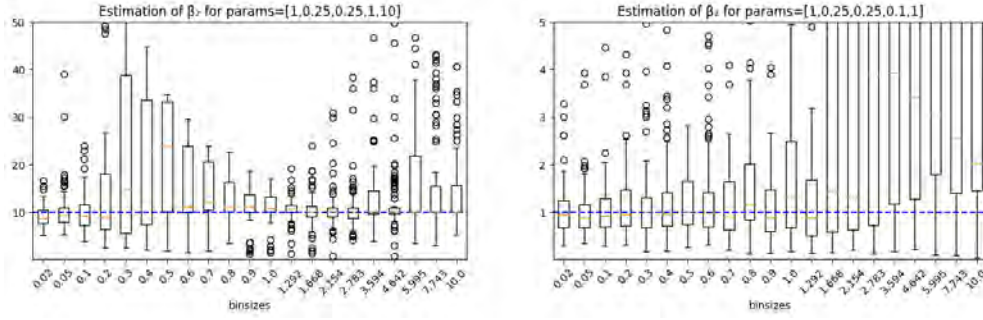


Figure 2.14: Sensitivity of  $\beta_2$  to binsizes

The preceding graphs show that the variance of the estimates increases significantly as the binsize exceeds the characteristic time  $\frac{1}{\beta}$ . For a process with a relatively large  $\beta_2$ , we observe a rapid deterioration in variance with increasing binsize, as expected.

### Sensitivity of the estimate to initial conditions

To study the sensitivity to initial conditions of the estimates made with the whittle method we simulated several hawkes processes with a two-exponential kernel with the following parameters:

$$\mu = 1, \alpha_1 = 0.2, \alpha_2 = 0.4, \beta_1 = 1, \beta_2 = 10$$

We also varied the size of the binsize among the following values: 0.02,0.05,0.1,0.2.

Then we estimated the parameters using the 'SLQP' optimization algorithm starting from initial conditions randomly generated according to a uniform distribution on an interval of values more or less distant from the values of the parameters from which the hawkes processes were simulated. These intervals model a degradation of the initial conditions. We have differentiated between positive degradation (when we start from values higher than the true values for the estimation) and positive degradation (when we start from values lower than the true values for the estimation) since we have noticed that the estimation does not behave symmetrically according to the sign of the degradation. The ranges of values retained for the initial conditions  $x_0$ , with  $x$  is the vector of the true values of the parameters, are the following :

- $] -200\%, -190\% ] : x_0 \sim U(\frac{x}{3}, \frac{x}{2}) ;$
- $] -100\%, -90\% ] : x_0 \sim U(\frac{x}{2}, \frac{x}{1.9}) ;$
- $] -50\%, -40\% ] : x_0 \sim U(\frac{x}{1.5}, \frac{x}{1.4}) ;$
- $] -20\%, -10\% ] : x_0 \sim U(\frac{x}{1.2}, \frac{x}{1.1}) ;$
- $] -5\%, 0\% ] : x_0 \sim U(\frac{x}{1.05}, x) ;$
- $] 0\%, 5\% ] : x_0 \sim U(x, 1.05x) ;$
- $] 10\%, 20\% ] : x_0 \sim U(1.1x, 1.2x) ;$
- $] 40\%, 50\% ] : x_0 \sim U(1.4x, 1.5x) ;$
- $] 90\%, 100\% ] : x_0 \sim U(1.9x, 2x) ;$
- $] 190\%, 200\% ] : x_0 \sim U(2.9x, 3x) ;$

To measure the impact of these degradations on the quality of the estimate, we used two metrics:

- Deviation of the estimate (which is the median of the different estimates) from the exccate value of the parameter
- The standard deviation of the different estimates

These two metrics have been computed for the 5 parameters ( $\mu, \alpha_1, \alpha_2, \beta_1, \beta_2$ ) of the two-exponential kernel from 75 estimates (each corresponding to a different simulation of a hawkes process for the same parameters).

The study showed that for acceptable binsizes and for initial conditions close to the exact values the whittle method allows to estimate the parameters well by using the median of the estimates. The method is no longer efficient when we move away from the true values for the initial conditions. For the binsize values used, chosen among the acceptable binsizes for the whittle estimation, the binsize does not have a significant impact on the quality of the estimation.

These results are illustrated by the following heatmaps:

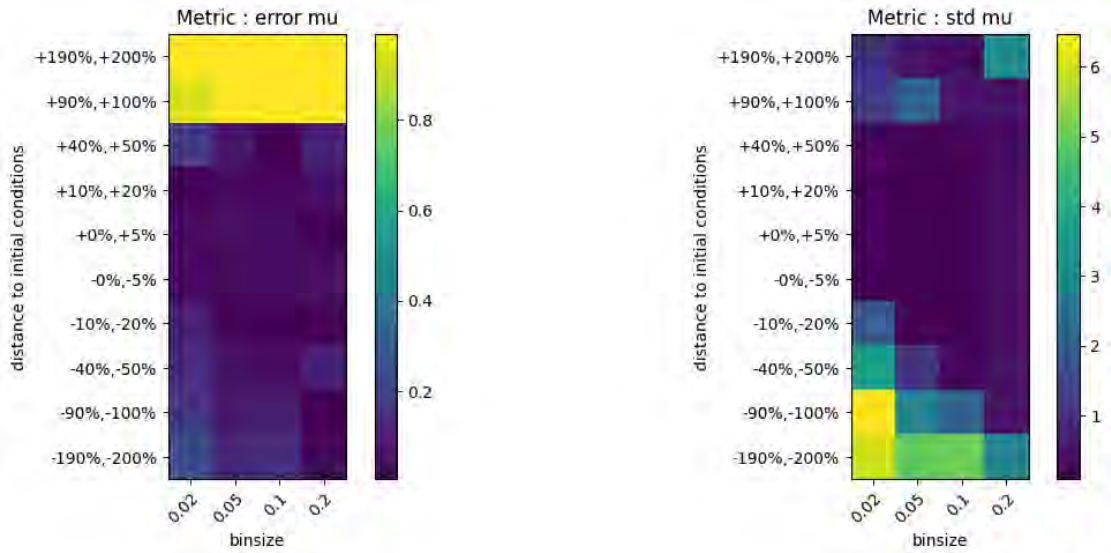


Figure 2.15: Sensitivity of  $\mu$  estimate to the variation of initial conditions

A large positive deviation of the initial conditions from the real values of the parameters is more detrimental to the estimation of  $\mu$  than a negative deviation. Then one might more want to start from a low value of the parameters to estimate  $\mu$ . The precision of the estimates is approximately the same regardless of the degradation of the initial conditions.

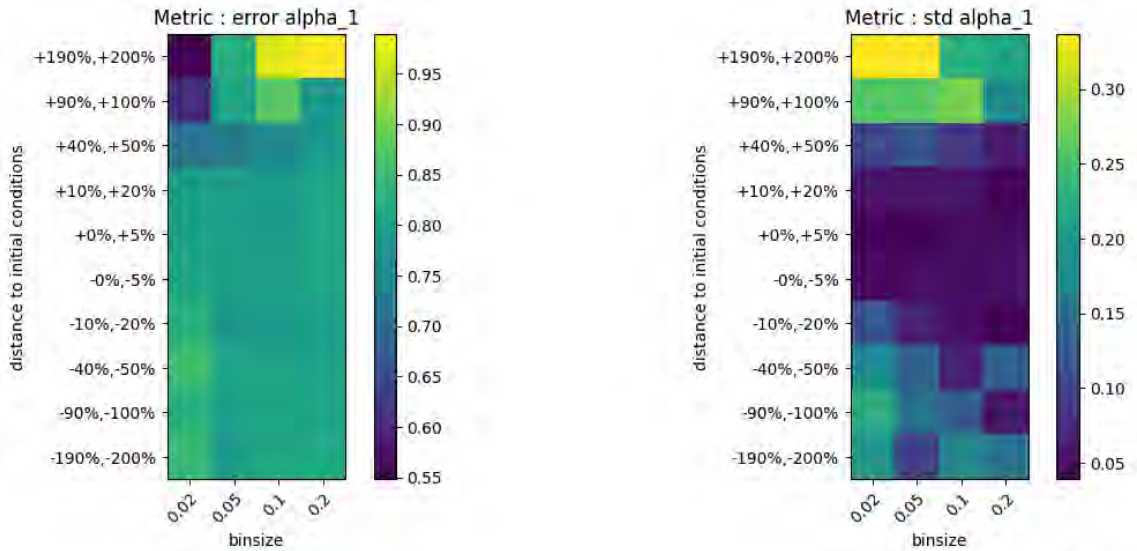


Figure 2.16: Sensitivity of  $\alpha_1$  estimate to the variation of initial conditions

The precision of the estimation of  $\alpha_1$  is better when the initial conditions are close to the real values of the parameters.

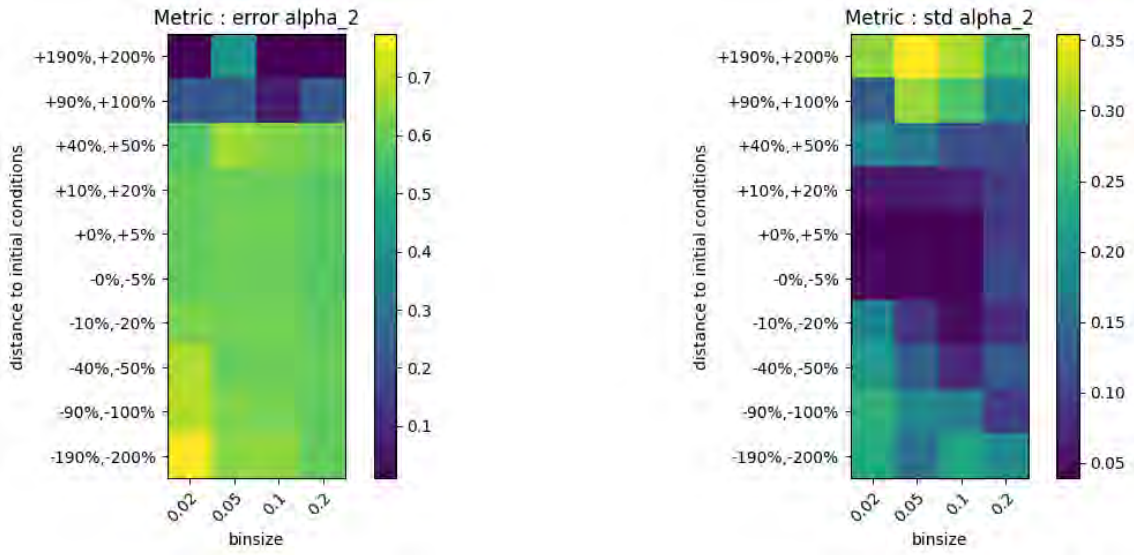


Figure 2.17: Sensitivity of  $\alpha_2$  estimate to the variation of initial conditions

The same phenomenon is observed for alpha 2.

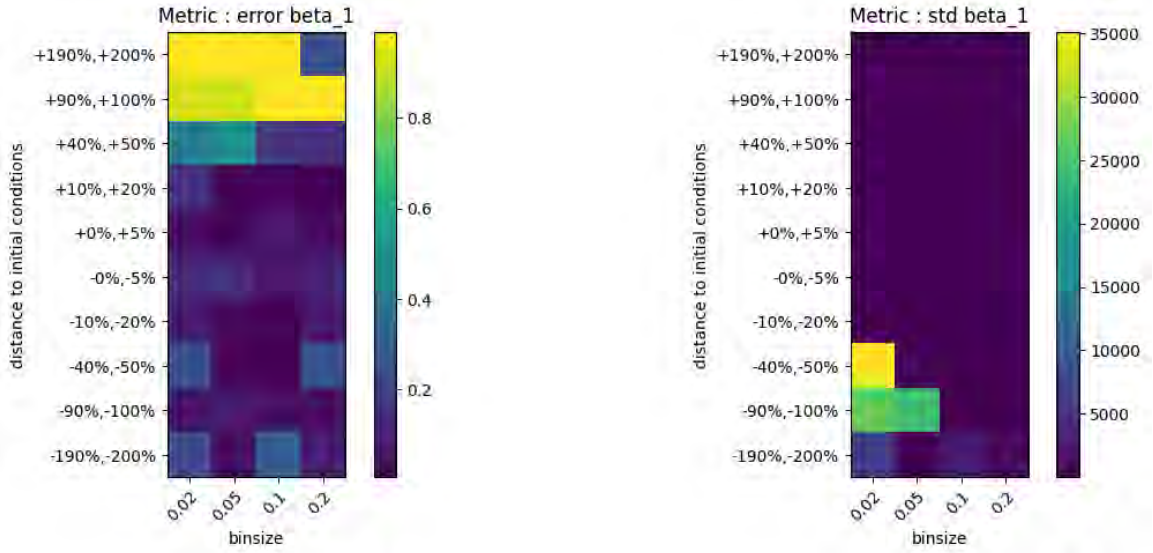


Figure 2.18: Sensitivity of  $\beta_1$  estimate to the variation of initial conditions

The same phenomenon as mu is observed for the estimate of beta 1 : a significant positive deviation of the initial conditions from the real values of the parameters harms the estimation more than a negative deviation. The precision of the estimates is also more or less the same regardless of the degradation of the initial conditions and is globally bad.

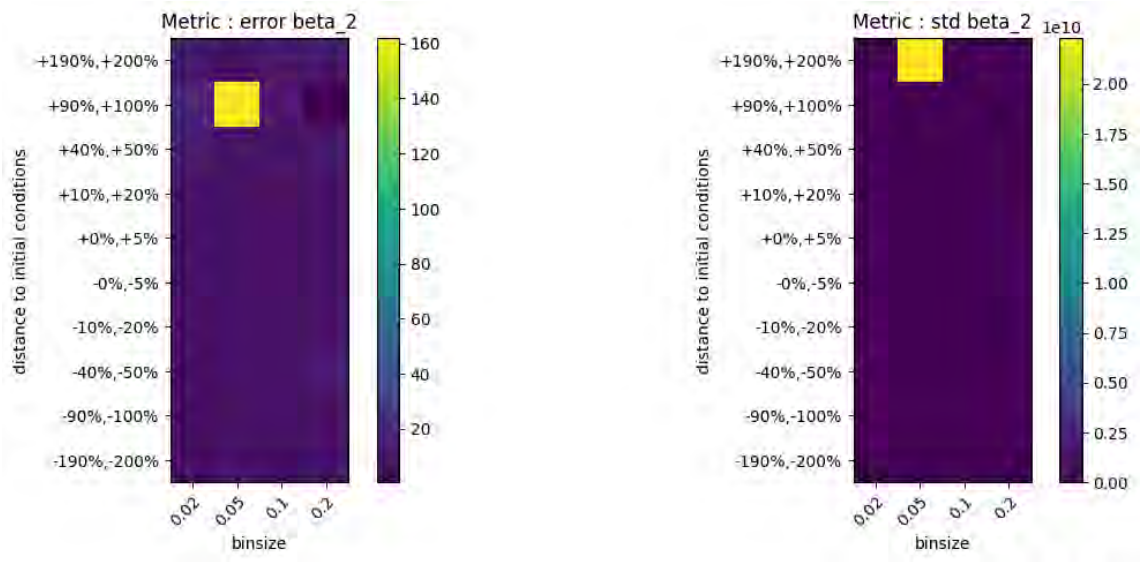


Figure 2.19: Sensitivity of  $\beta_2$  estimate to the variation of initial conditions

The estimates of beta 2 are even less precise, the standard deviation is around  $10^{10}$ .

### 2.3.2 Comparing Whittle and Unif in the two dimensional case

In this graph, we display boxplots representing the estimations of various parameters using both the Whittle and Unif methods. It is important to note that the true parameters we used are as follows:  $\mu = 1$ ,  $\alpha_1 = 0.2$ ,  $\alpha_2 = 0.4$ ,  $\beta_1 = 1$ , and  $\beta_2 = 10$ . The estimations shown were obtained when starting from the exact initial conditions.

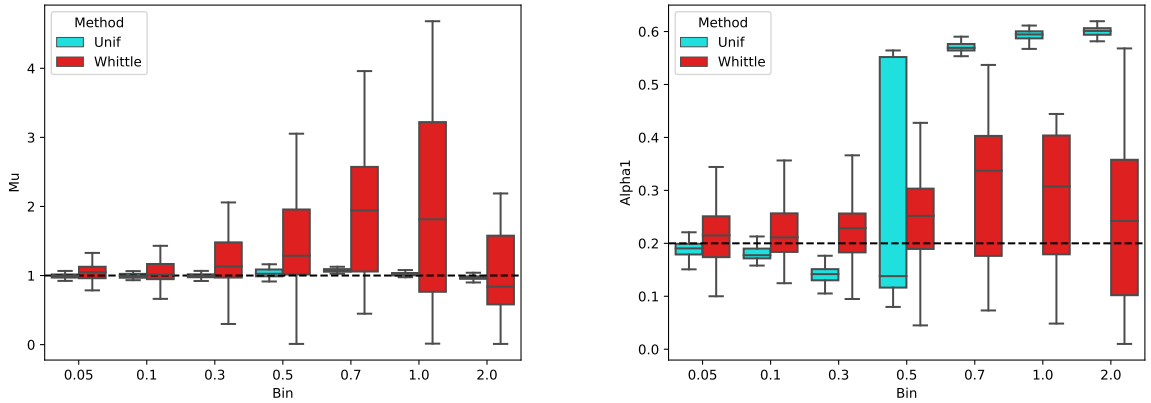


Figure 2.20: The estimation of  $\alpha_1$  for different bin widths

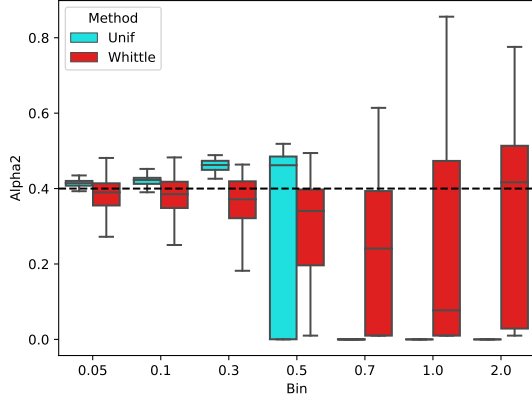


Figure 2.21: The estimation of  $\alpha_2$  for different bin widths

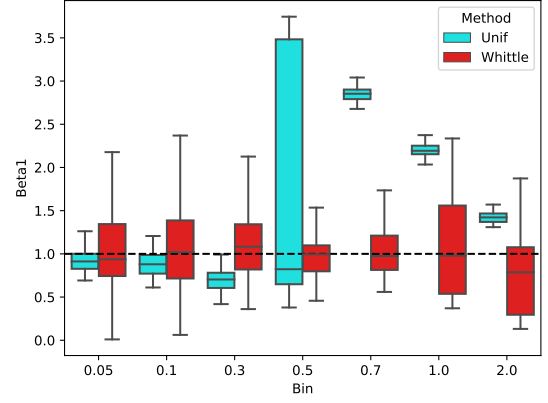


Figure 2.22: The estimation of  $\beta_1$  for different bin widths

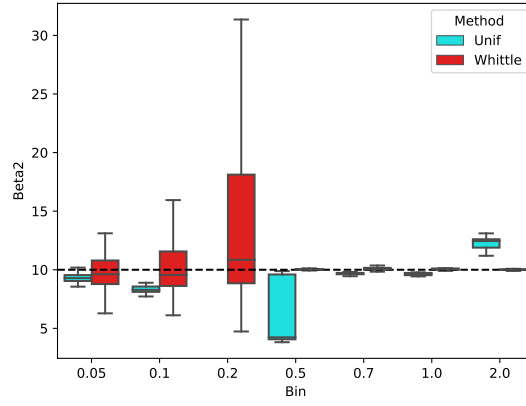


Figure 2.23: The estimation of  $\beta_2$  for different bin widths

From the various graphs, it is evident that the Unif method outperforms the Whittle estimation in terms of accuracy when dealing with very small bin sizes. Additionally, the Unif method exhibits lower variance compared to the Whittle estimation. In contrast, the Whittle estimation does not tend to treat the process as a one-exponential Hawkes process when the number of elements in the bins exceeds 2. However, it is important to note that the results obtained for  $\alpha_2$  with a bin size greater than 0.5 are notably poor, characterized by large variances.

## Chapter 3

# Supervised Learning

Supervised Learning offers a promising approach for our estimation task. It possesses the advantage of being less susceptible to the influence of data binsize, as highlighted by Tom Keane [3]. Consequently, we can speculate that this method has the potential to outperform other approaches, including MLE\_random, particularly when dealing with larger bin sizes.

The aim of this research is to develop a method that accurately predicts both the branching ratio ( $\eta$ ) and baseline intensity ( $\mu$ ) as a regression task using neural networks. In our research, we developed an approach for estimating  $\mu$ ,  $\alpha$ , and  $\beta$  using a supervised method. However, this initial attempt was unsuccessful. Subsequently, we drew inspiration from the approach proposed in the referenced paper to refine our methodology. It is important to note that in this section the conditional intensity is defined the following way to comply with the notations used in [3]:  $\lambda(t) = \mu + \sum_{ti < t} \alpha \beta e^{-\beta(t-t_i)}$ . Thus the branching ration is  $\eta = \frac{\alpha}{\beta}$ .

### 3.1 Supervised Method

We hypothesize that this can be achieved by leveraging the incremental count data  $x_i = \{N_j^\Delta\}_{j=1, \dots, T/\Delta}$ , where  $T$  is the horizon and  $\Delta$  is the discretization step. In order to evaluate the performance of our method, we generated 100,000 training data processes based on the approach outlined. The test data points  $\{y_i\}_{i=1, \dots, m}$  are realizations of Hawkes processes with the same conditional intensity function parameters.

#### 3.1.1 Model Architecture

For our neural network model, we employed a feed-forward Multilayer Perceptron (MLP) with an input dimension of  $T/\Delta$ . The network consists of six hidden layers, each containing  $T$  neurons. The output layer consists of two neurons representing the parameters  $\theta = \{\eta, \mu\}$ . ReLU activation function was used for the hidden layers, and L2 regularization with weight 0.001 was applied. The final output layer used the linear activation function.

While the output range is strictly non-negative, the inclusion of a ReLU activation function caused performance issues. This general network structure was designed to enable the unsupervised deployment of a supervised learning algorithm. Its goal was to ensure that learning is not restricted by neuron capacity. In case the capacity is too large, the regularization term down-weights unnecessary neurons. Although this design may not be optimal for all applications, it provides sufficient flexibility for most scenarios. The network was trained for a maximum of 500 epochs, and early stopping was defined as a reduction in validation loss of less than 0.01 over 25 epochs. This criterion allowed most applications to fully train within 200 epochs. In practice, the model was trained in less than ten minutes using 100,000 training samples. The training time is relatively short, and the test time is nearly instantaneous, unlike some other methods (e.g., the uniform method) which is an advantage.

#### 3.1.2 Different Learning Method

We explored two different learning methods: The first one using  $m$  disjoint observations of length  $T/\Delta$ , and the other one using  $m$  segments of a single observation of length  $mT/\Delta$ . The first method

allows us to study the entire process, while the second method focuses only on the stationary part of the process. In our work, we decided to choose the first method for our analysis.

### 3.1.3 Data Generation

To align with the chosen estimation method and our objective of estimating  $\eta$  and  $\mu$ , we employed a specific data generation process.

To generate  $n$  training processes,  $x_{i\{i=1,\dots,n\}}$ , with an expected activity level of  $E$ , the following steps were undertaken:

1. Sample random normal variables,  $\varepsilon_i \sim N(E, \sigma^2)$ , for  $i = 1, \dots, n$ .
2. Sample  $\eta_i \sim U(a, b)$ , where  $0 < a < b < 1$  are predetermined based on prior beliefs.
3. Calculate  $\mu_i = \varepsilon_i \cdot (1 - \eta_i) \cdot T$ .
4. Sample  $\beta_i \sim U(p, q)$ , where  $0 < p < q$  are predetermined based on prior beliefs.
5. Calculate  $\alpha_i = \beta_i \cdot \eta_i$ .
6. Generate the  $i$ th training process using an appropriate method and corresponding parameter set:  $\alpha_i, \beta_i, \mu_i$ .
7. Discretize the event times at intervals of length  $\Delta$  to form the respective incremental count processes  $x_i = \{N_j^\Delta\}_{j=1,\dots,T/\Delta}$ .

It is worth noting that the parameters  $\eta$  and  $\beta$  in the training process are uniformly distributed. This consistent data generation approach will also be employed for the test process to validate our model.

### 3.1.4 Estimation of $\alpha$ and $\beta$

Initially, we will present the method proposed by the article and subsequently demonstrate its validity. However, a significant challenge arises with this method: it necessitates the storage of both the parameters and the discretized processes of the training data (i.e.,  $x_{i\{i=1,\dots,n\}}$ ) in memory. Unfortunately, in practical scenarios, this requirement poses a major limitation.

Given  $m$  discretised test processes,  $\{y_i\}_{i=1,\dots,m}$ , with a time horizon  $T$  and a discretisation step size of  $\Delta$ , the following steps were taken to estimate  $\alpha$  and  $\beta$ :

1. Using an estimate of  $\eta$ ,  $\hat{\eta}$ , generate training processes,  $\{x_i\}_{i=1,\dots,n}$ , using the method outlined in Section 4.1.1 with  $\eta_i \sim U(\hat{\eta} - \delta_\eta, \hat{\eta} + \delta_\eta)$ . Note:  $\delta_\eta$  is chosen by the user based on confidence in  $\hat{\eta}$ .
2. For  $k = 1, \dots, K$ , calculate:

$$s_k = \text{mean}_i(\{\max(x_{ij}) \mid \alpha_i \in ((k-1)\delta_\alpha, k\delta_\alpha)\})$$

where  $K = \lceil \max_i(\alpha_i)/\delta_\alpha \rceil$  and  $\delta_\alpha$  is chosen based on the level of variability in the data.

3. Fit a normal linear model to the response  $(k - 0.5)\delta_\alpha$  using  $s_k$  as the independent variable.
4. Calculate the median value of  $\max_j(y_{ij})$ .
5. Using the fitted linear model, predict the corresponding value of  $\hat{\alpha}$ .
6. Calculate  $\hat{\beta} = \hat{\alpha}$ .

The foundation of this method rests on the assumption that for discrete Hawkes processes with similar values of  $\mu$ ,  $\beta$ , and  $\alpha$  within a specific interval, a linear regression exists between  $s_k$  and  $k$ . However, the paper does not provide a formal proof of this relationship. Therefore, our objective is to empirically show this correlation.

We chose to conduct a test by dividing the interval of  $\alpha$  into 100 smaller intervals, denoted as  $[0.2, 0.6]$ . Let  $K$  represent the total number of intervals and  $M$  denote the number of simulated processes per interval. For each interval  $k$  from 1 to  $K$ , we simulate  $m$  Hawkes processes, where  $\alpha$  is uniformly selected from within the corresponding smaller interval.

Subsequently, we compute  $s_k$  for all  $k$  and here is the linear regression.

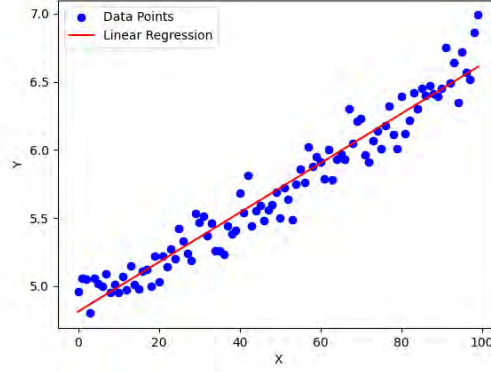


Figure 3.1: Expression of  $s_k$  as a function of  $k$

According to this graph, we can affirm that the method used to estimate  $\alpha$  and  $\beta$  seems legitimate. However, as good as this fit may seem ( $R^2 = 0.94$ ), the degradation on the estimation of  $\alpha$  and thus on  $\beta$  is very important and makes this method unusable to estimate these two parameters. For this reason, we will mainly concentrate on the estimates of  $\mu$  and  $\eta$  in what follows. Surprisingly, in an attempt to train a neural network to estimate  $\alpha$  from  $\mu$ ,  $\eta$  and the maximum non empty bin, the results were bad, as if the NN wasn't capable to capture this linear relationship.

### 3.1.5 Training and Evaluation

In order to create a comprehensive model, it was crucial to include a wide range of parameters in the training data. Consequently, we carefully chose the parameter ranges for  $\alpha$  as  $[0.05, 0.8]$  and for  $\beta$  as  $[1, 3]$ . These ranges were selected to cover a diverse set of values and ensure the model's effectiveness.

When evaluating the model using test data, we will employ parameter intervals derived from these ranges. It is important to note that if the range of parameters in the test data falls outside the training data, the model's performance may be adversely affected. Therefore, it is essential to ensure that the test data parameters are within the range of parameters encountered during training to achieve optimal results.

In our analysis and corresponding graphs, we aim to demonstrate the influence of binsize on the estimation process. Additionally, we seek to observe the impact of the selected parameters on the estimation outcomes. By examining these factors, we can gain insights into how varying binsize affects the accuracy of the estimation, as well as how different parameter choices impact the overall estimation results.

### 3.1.6 Results

#### 3.1.6.1 Influence of $\eta$ range

We start by analyzing the influence of the range of the branching ratio on the performance of the model. To do this, the model was trained and tested on three different ranges for  $\eta$ :  $[0.5, 0.8]$ ,  $[0.3, 0.8]$  and  $[0.05, 0.8]$  and compared to the **MLE\_random** method which was identified as the most promising for binned Hawkes precesses in chapter 1. These were the results we found:

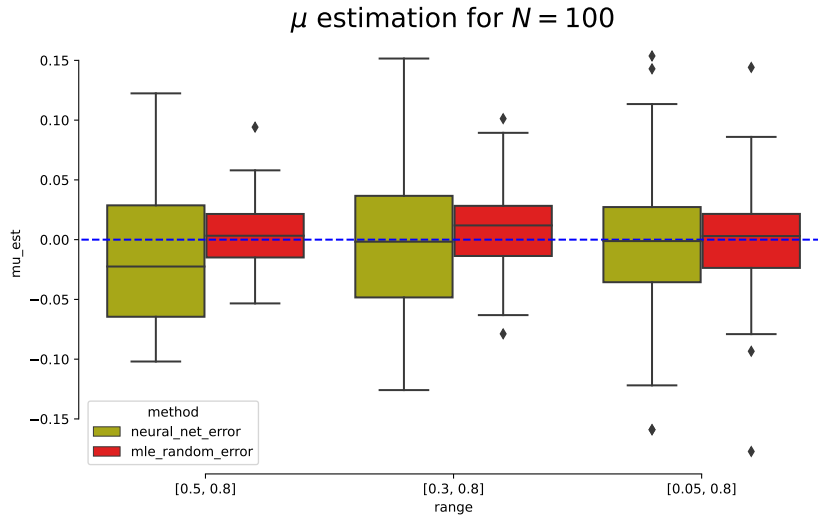


Figure 3.2: The estimation of  $\mu$  for different  $\eta$  ranges

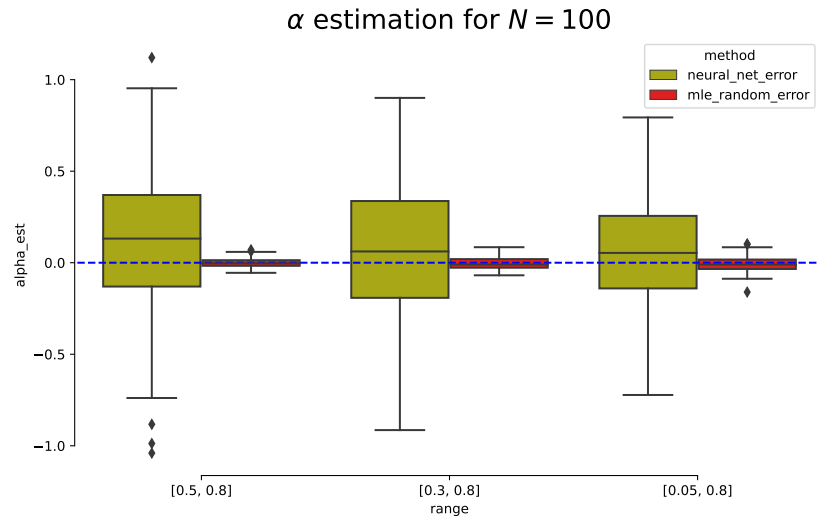


Figure 3.3: The estimation of  $\eta$  for different  $\eta$  ranges

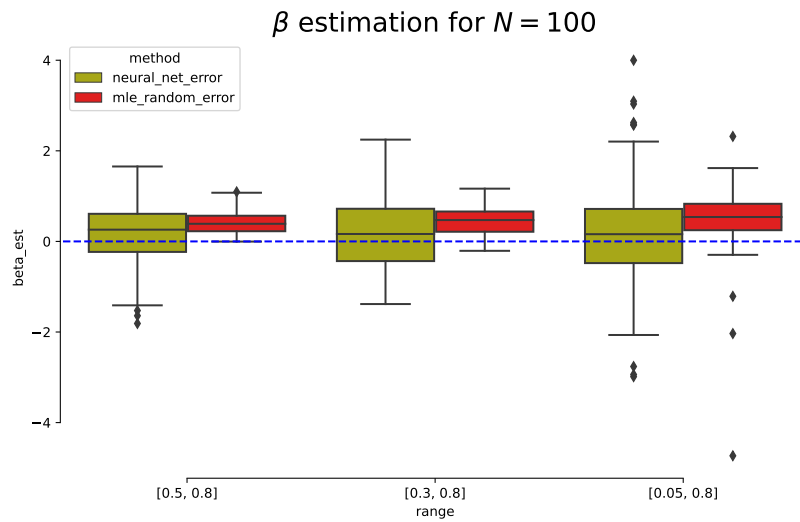


Figure 3.4: The estimation of  $\beta$  for different  $\eta$  ranges

We observe that the size of the interval for  $\eta$  on which the neural network is trained has no influence on its performance. Similar results were found in [3] to the only difference that, surprisingly, the boxplots we found for the  $\mu$  estimation are much thinner while the methods used were exactly the same. Also note that the **MLE\_random** method is biased for the  $\beta$  estimation which is normal due to a big binsize as discussed in chapter 1 while the neural network error isn't biased. Thus, the neural network could be an interesting substitute to **MLE\_random** in cases of large binsizes to estimate  $\beta$ , in the same way as the Whittle method. Given these observations, in all the following graphs, the neural network will be trained on the largest  $\eta$  range  $[0.05, 0.8]$ .

### 3.1.6.2 Comparing all methods

We compare the performance of the neural network with all the other methods studied in chapter 1. Recall that the boxplots of chapter 1 were plotted for 100 estimations on simulated Hawkes processes of parameters  $\mu = 1.0, \alpha = 0.5$  and  $\beta = 1.0$ . However, since the neural network was trained on the ranges  $\eta \in [0.05, 0.8]$ ,  $\beta \in [1, 3]$  and more or less  $\mu \in [0.2, 0.95]$  (see more details in the code), the boxplots showed below for this method we performed for simulated Hawkes processes with different parameters within these training ranges. Thus, to compare similar performance across all methods, the binsizes used to train the neural network were chosen in order to match the mean number of events per non empty bin, which is the key element of precision for binned estimation methods. The graphs show the boxplots of the relative error to the true parameters.

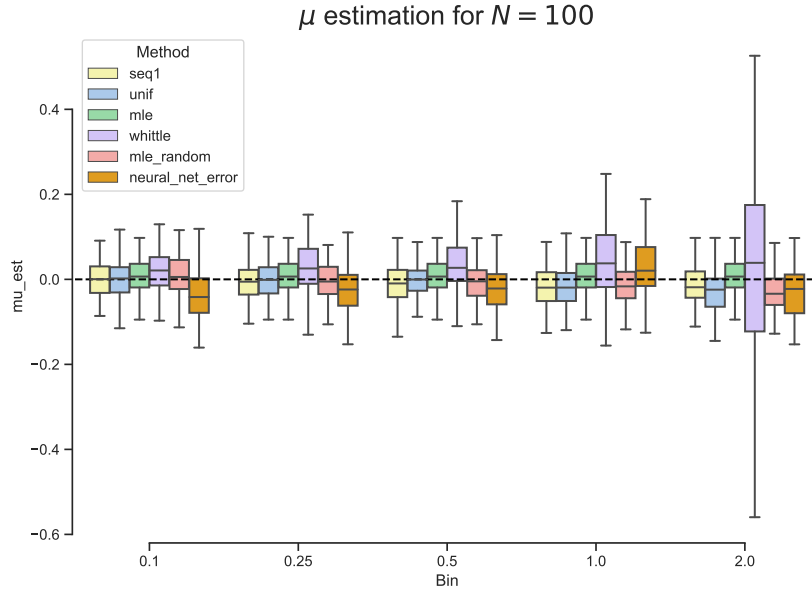


Figure 3.5: The estimation of  $\mu$  for different binsizes

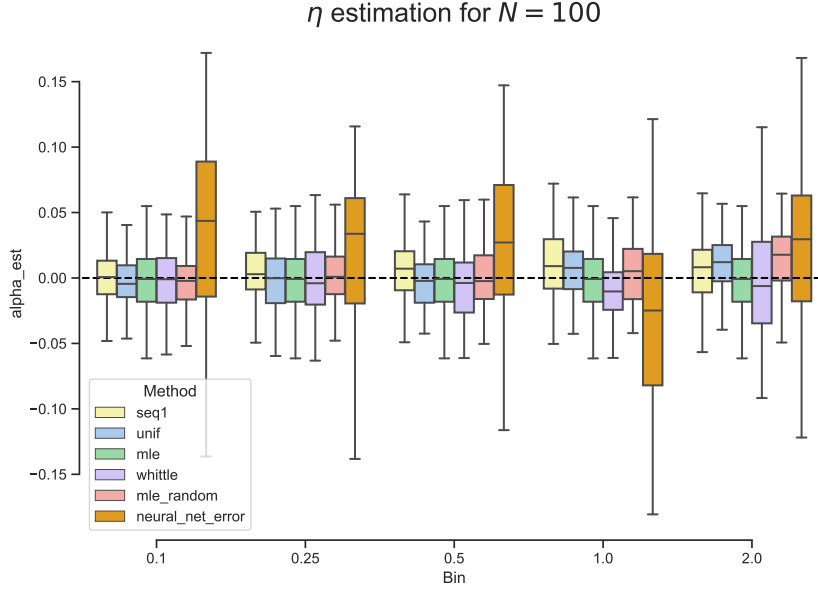


Figure 3.6: The estimation of  $\eta$  for different binsizes

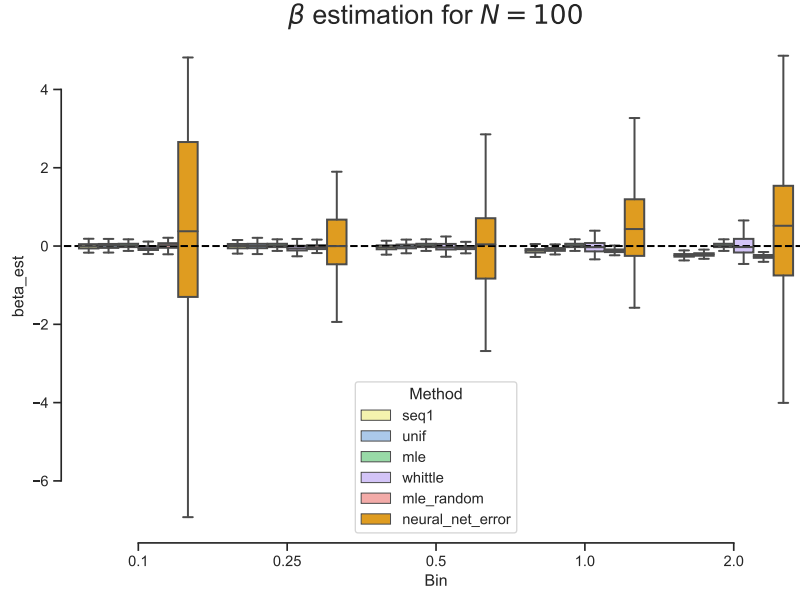


Figure 3.7: The estimation of  $\beta$  for different binsizes

We observe that the neural network has good performance for estimating  $\mu$ : the variance of the boxplots are close to the **MLE\_random** method but slightly biased. However, precision degrades for the branching ratio  $\eta$  where the variance becomes high. For  $\beta$ , the estimation performs very badly. A very interesting property though is that the NN doesn't seem affected by the bin size which makes it robust to this feature on which the other methods are very dependent.

Thus, the neural network performs worse than **MLE\_random** for  $\mu$  and  $\eta$  no matter the bin size and doesn't seem relevant to solve the bias problem that appears for  $\beta$  when the bin becomes large, the Whittle method being much more precise. Thus, unfortunately, the machine learning method we exposed here doesn't seem to have advantages to estimate binned Hawkes parameters. In an attempt to improve the precision on  $\beta$ , a neural network was trained to estimate this parameter given  $\mu$ ,  $\eta$  and the maximum non empty bin but then again, results were bad.

### 3.1.6.3 Improving the neural network performance: the LSTM network

The Long Short Term Memory networks can turn in very useful to analyze complex time series with seasonality and trends because they are capable to capture both long and short term memory patterns by backpropagating on single and sequences of neurons. The size of this sequence can be adjusted by the *timesteps* parameter in the Keras library in Python. In this section we determine by cross-validation which choice of the *timesteps* parameter is the most efficient to estimate  $\mu$  and  $\eta$ . In the legend of the graphs, the method labeled by *lstm\_tps\_50* means the size of the sequence is  $100/50 = 2$ . Thus, for *lstm\_tps\_25*, the size of the sequence is 4.

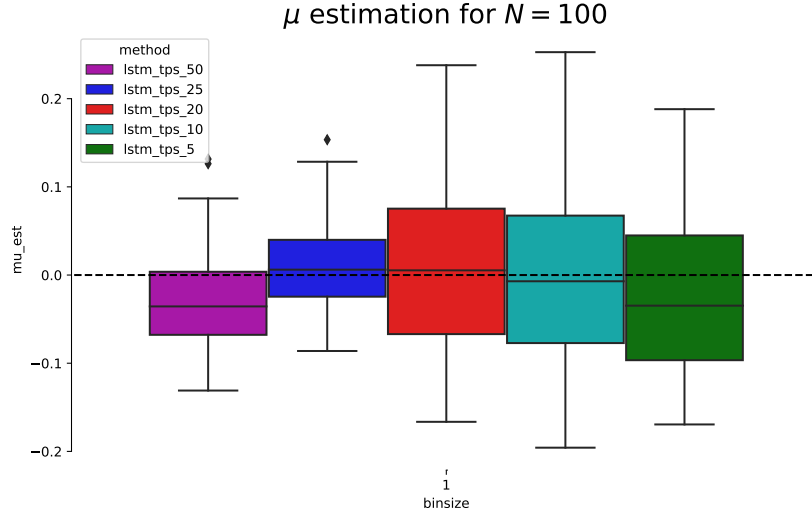


Figure 3.8: The estimation of  $\mu$

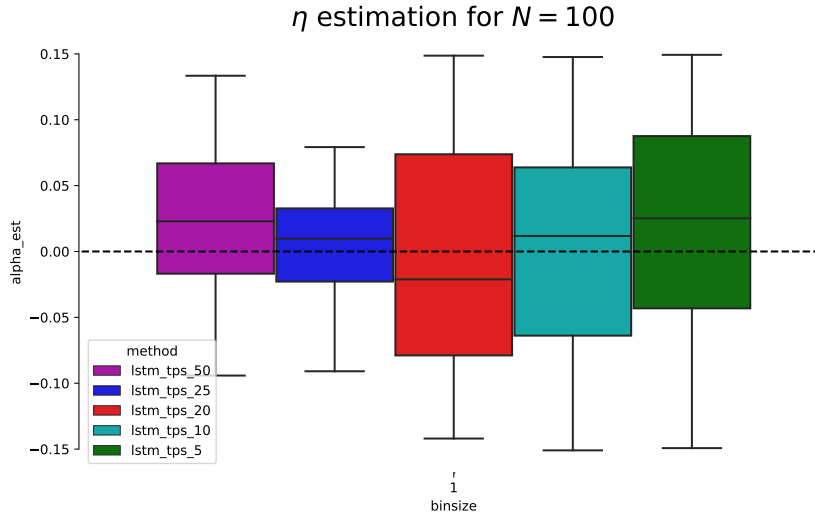


Figure 3.9: The estimation of  $\eta$

It seems clear that the 4-sized sequence is the most efficient to estimate both  $\mu$  and  $\eta$  and has very little error. We will keep this choice in the following graphs. Next, we see how LSTM outperforms the traditional neural network to estimate  $\mu$  and  $\eta$ .

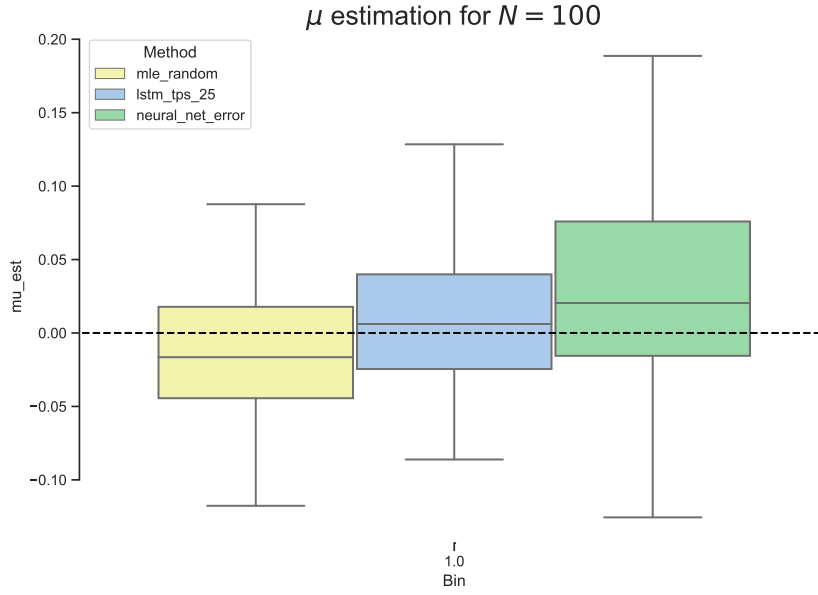


Figure 3.10: The estimation of  $\mu$  for different timesteps

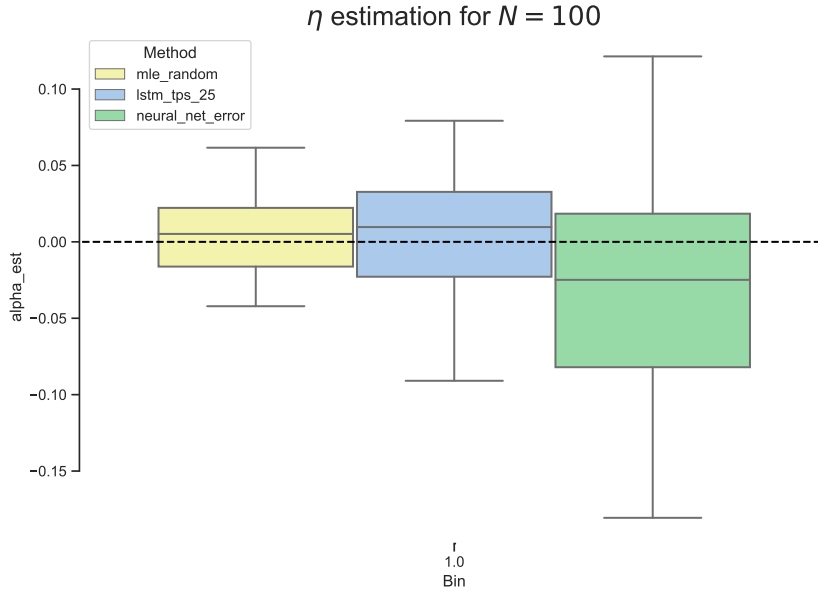


Figure 3.11: The estimation of  $\eta$  for different timesteps

These graphs show that LSTM networks seem very promising, performing almost as good as **MLE\_random**. Maybe the performance could be increased by using a dynamic size of the timestep that would vary according to the distribution of non empty bins to capture more precisely local excitation. Many avenues are still to explore to try to improve the performance of machine learning methods for binned Hawkes process.

To conclude, machine learning methods seem interesting to estimate binned Hawkes processes. Estimates can be done quickly and accurately once the model trained. However, many hyperparameters need to be adjusted which can be time consuming. Using ML techniques also requires using data formatting subject to several constraints which is a disadvantage. All in all, as much as this technique seems interesting, it seems difficult to outperform **MLE\_random** that estimates accurately, quickly and with no particular constraints.

## Chapter 4

# Fitting Real Financial Data

In our study, we obtained a substantial dataset comprising thirty days' worth of financial data, each day represented by a separate file. This dataset captured the diverse range of orders observed throughout a typical trading day. It is worth noting that the smallest time scale captured in our data was at the microsecond level, ensuring that orders were recorded accurately, with no overlapping orders occurring simultaneously. Our primary objective was to assess the effectiveness of our tested method, which was initially validated using degraded simulated data. Now, we sought to apply this method to the real financial data we collected, specifically aiming to determine if it yielded consistent results and successfully captured the intricacies present in actual market dynamics. By subjecting this extensive dataset to our method and analyzing the outcomes, we aimed to gain insights into the compatibility and efficacy of our approach when dealing with complex financial data.

In order to assess the suitability of our Hawkes process model for fitting the data, we focused on studying the integrated timestamps, which were expected to follow an exponential distribution.

Say  $\{t_1, t_2, \dots, t_k\}$  is a realization over time  $[0, T]$  from a point process with conditional intensity function  $\lambda^*(\cdot)$ . If  $\lambda^*(\cdot)$  is positive over  $[0, T]$  and  $\Lambda(T) < \infty$  a.s., then the transformed points  $\{\Lambda(t_1), \Lambda(t_2), \dots, \Lambda(t_k)\}$  form a Poisson process with unit rate according to [5].

For testing whether a series of points form a Poisson process (see Cox and Lewis 1966 for an extensive treatment) we check the interarrival times,  $\{\tau_1, \tau_2, \tau_3, \dots\} = \{t_1^*, t_2^* - t_1^*, t_3^* - t_2^*, \dots\}$ , i.i.d., to ensure  $\tau \sim \text{Exp}(1)$ . A qualitative approach is to create a Q-Q plot for  $\tau_i$  using the exponential distribution (see e.g., Figure 5.2a). Otherwise, a quantitative alternative is to run Kolmogorov-Smirnov test.

In conclusion, we will examine whether  $\{\Lambda(t_1), \Lambda(t_2) - \Lambda(t_1), \dots, \Lambda(t_k) - \Lambda(t_{k-1})\} \sim \text{Exp}(1)$ .

### 4.1 Results for the MLE method

Throughout our analysis, we observed that the MLE method emerged as the superior approach. Our objective now shifts towards investigating the feasibility of modeling the non-degraded data using Hawkes processes. Furthermore, we aim to determine the optimal number of exponential components required to achieve the best possible fit to the data. We want to gain insights into the capability of Hawkes processes when applied to real-world financial datasets.

Our methodology involves conducting extensive tests on multiple days, employing a graphical approach to determine the optimal number of exponential components. Specifically, we aim to model an hour of trade using the Hawkes process. Notably, we observed that the fitting results across different days exhibit considerable similarity. As a result, we present the graphical representations solely for a single day, highlighting its representative nature for the overall analysis. By focusing on this particular day, we can effectively showcase the patterns and outcomes derived from our modeling approach.

An Hawkes process with an exponential kernel appears to be a suitable model for fitting financial data. However, determining the appropriate number of exponentials to consider is crucial. As anticipated, the fit tends to improve with an increasing number of exponentials. Notably, there is a noticeable progression in the quality of the fit between one, two, and three exponentials. The addition of a fourth exponential further enhances the fit, although the improvement from including a fifth exponential is comparable to that of four exponentials. Consequently, it can be observed

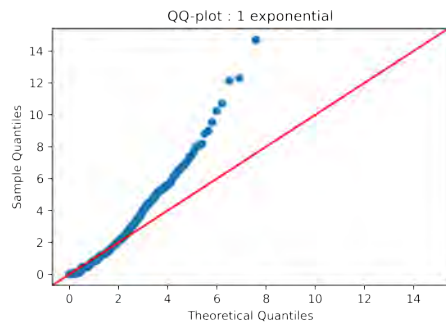


Figure 4.1: QQ-Plot: 1 exponential

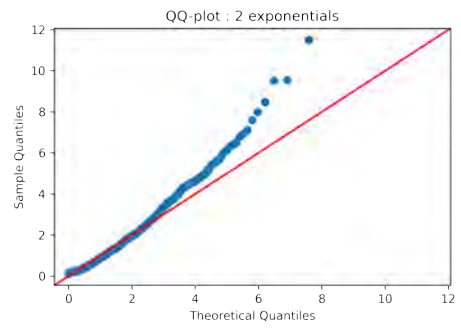


Figure 4.2: QQ-Plot: 2 exponentials

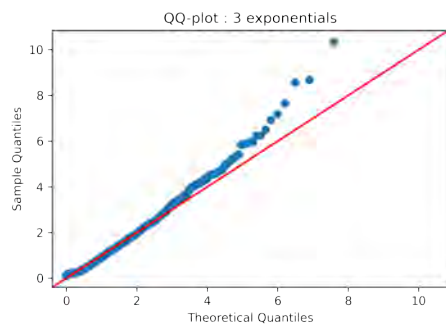


Figure 4.3: QQ-Plot: 3 exponentials

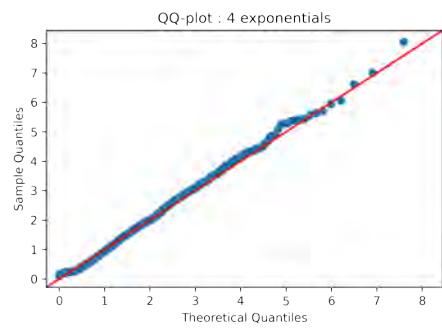


Figure 4.4: QQ-Plot: 4 exponentials

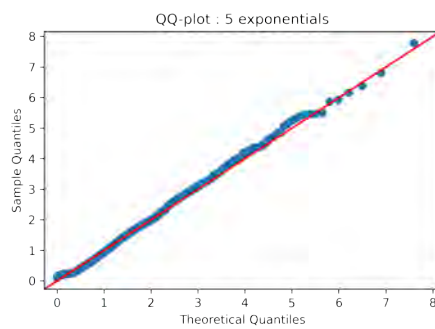


Figure 4.5: QQ-Plot: 5 exponentials

Figure 4.6: Multiple Images with Titles

that employing three or four exponentials is sufficient for the modelization. However, it is essential to consider the computational capacity when making this choice.

## 4.2 Results with MLE\_random with degraded data

In this section, we focused on degrading the previously used data from the previous section. The degradation process involved reducing the precision of the data from  $10^{-6}$  seconds to  $10^{-5}$ ,  $10^{-4}$ , and so on. The primary objective of this part was to fit a four-exponential model to the degraded data and determine the acceptable level of degradation for the new estimation method. By examining the performance and accuracy of the model under different levels of degradation, we aimed to identify the threshold beyond which the estimation method's results become unreliable or significantly affected. This analysis allowed us to gain insights into the robustness and limitations of the estimation method when applied to degraded financial data.

With a binsize smaller than  $10^{-6}$ , the kernel is effectively blocked.

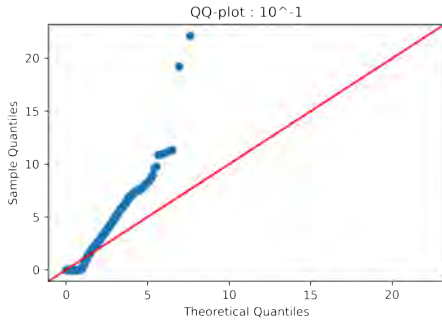


Figure 4.7: QQ-Plot: Bin =  $10^{-1}$

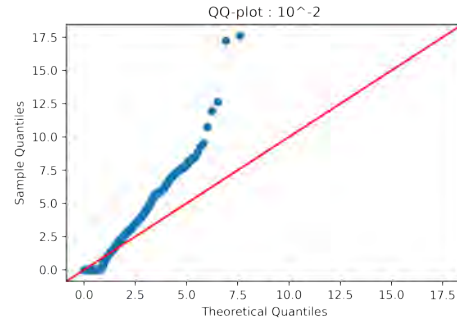


Figure 4.8: QQ-Plot: Bin =  $10^{-2}$

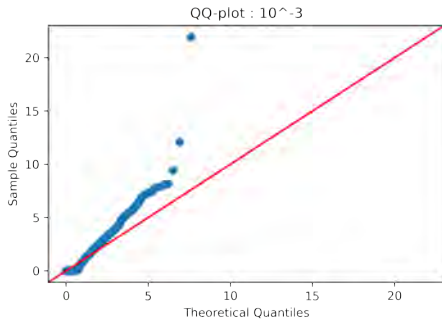


Figure 4.9: QQ-Plot: Bin =  $10^{-3}$

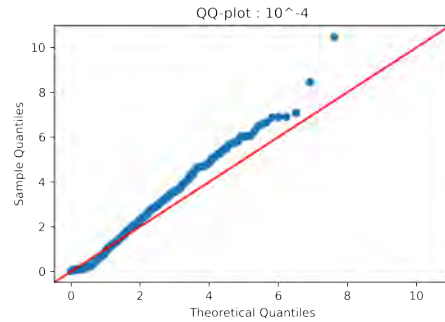


Figure 4.10: QQ-Plot: Bin =  $10^{-4}$

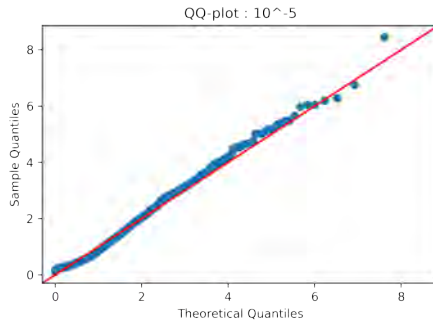


Figure 4.11: QQ-Plot: Bin =  $10^{-5}$

When the data is degraded, it is expected that the fit and the method's performance would be adversely affected. As observed, the method becomes less usable for data with a precision exceeding the millisecond range. However, for bin sizes that are more precise, the method exhibits promising

results. These outcomes provide insights into the feasibility of employing the unif method with multiple exponentials on degraded data. It is crucial to note that the bin size should be smaller than a millisecond for accurate results. Nevertheless, it is important to consider that reducing the bin size to such a small value significantly increases the computational time required, ultimately diminishing the unif method's practical utility.

#### 4.2.1 Whittle estimation

To run the whittle estimation with the real financial data, it is needed to do it with a time window and a binsize well chosen. Par ailleurs, on remarque que concernant l'estimation de Whittle, la médiane de plusieurs estimations indépendantes entre elles apparaît comme un bon estimateur (voire les heat map) en pièce jointe. Pour respecter l'indépendance entre les estimations, on les réalise sur plusieurs jours de différence en supposant qu'ils n'ont pas d'influence les uns sur les autres. Le nettoyage des données se fait en éliminant les journées qui ne sont pas des courbes en U, caractéristique typique de l'activité financière journalière.

##### 4.2.1.1 Choice of a time window

When it comes to choosing a time window, there's a trade-off.

1. This period also needs to be fairly long, as the estimate requires a number of points.
2. It is necessary to find a fairly short period of the day during which financial activity is relatively stable, so that a single Hawkes process can be estimated..

With regard to the second property, we need to set a threshold for the number of points we need to estimate. We set this threshold at 1000 points, because estimates made with such a large number of points start to become satisfactory. As a result, for the average of the files, windowing with a period greater than 30 minutes is required.

For the second, we looked at changes in the average number of timestamps per binsize. We can see that the most stable area of the day is between 10am and 11am.

In conclusion, we choose a window [10am,11am] which is compatible with the first constraint.

##### 4.2.1.2 Choice of the bin size

According to our findings, the bin size should be chosen smaller than  $\frac{1}{\beta_{max}}$ . According to the MLE estimations,  $\beta_{max}$  is around 100000. As a result, we would have to take a binsize smaller than  $10^5$ . In practice, however, estimates are likely to take too long. This is why the following estimates are made using  $10^{-3}$ .

Let recall that the number of bins used for estimation ( $N$ ), the binsize ( $\Delta$ ) and the estimation horizon  $T$  are linked by the following relationship:

- $N = \frac{T}{\Delta}$

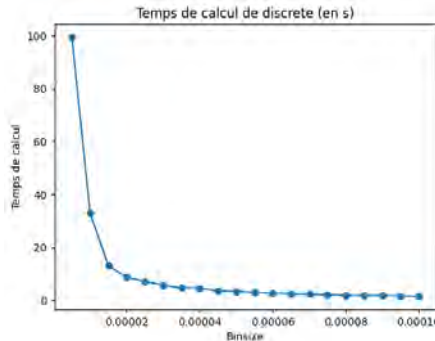


Figure 4.12: Influence of  $\Delta$  on the calculation time of the discrete function

The figure above shows the calculation time of the function that rearranges the transactions in the binsizes used in the whittle method, as a function of the binsize  $\Delta$ . The appearance of

the curve seems consistent with the fact that the number of bins is inversely proportional to the binsize.

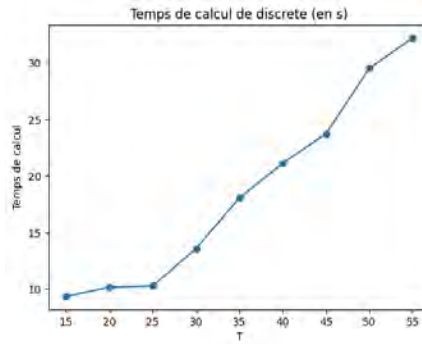


Figure 4.13: Influence of  $T$  on the calculation time of the discrete function

The figure above shows the calculation time of the function that rearranges the transactions in the binsizes used in the whittle method, as a function of the time horizon  $T$ . The appearance of the curve seems consistent with the fact that the number of bins is proportional to the time horizon .

#### 4.2.1.3 Results

To assess the feasibility of applying the Whittle estimation method to real financial data, we conducted fitting exercises on time series with intervals of 1 hour and 30 minutes. Specifically, we focused on two distinct time frames: either from 11:00 am to 12:00 pm or from 11:00 am to 11:30 am.

We utilized a two exponential Hawkes process. We reduced the data resolution to a finer scale of  $10^{-3}$ . This decision aimed to strike a compromise that allowed us to retain sufficient accuracy while optimizing the efficiency of the computational process. This procedure was repeated across multiple days. While the resulting plots displayed a good fit between the estimated models and the observed data (see the QQ-Plot), it is worth noting that there were instances where the plot, indicated deviations. However, it is important to acknowledge that the Whittle estimation method is noticeably influenced by the initial conditions set by the MLE method. This raises questions about the overall usefulness and applicability of the Whittle estimation method in scenarios where the initial conditions may not be reliably estimated.

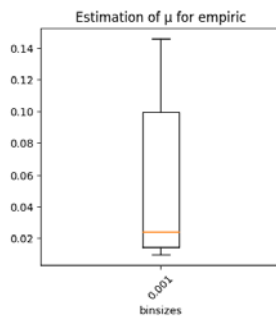


Figure 4.14

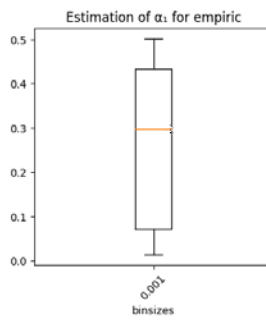


Figure 4.15

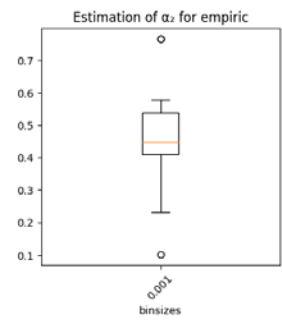


Figure 4.16

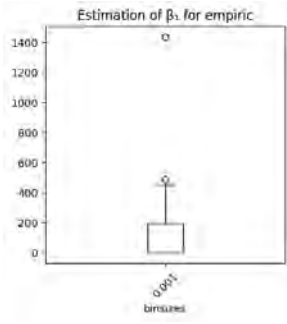


Figure 4.17

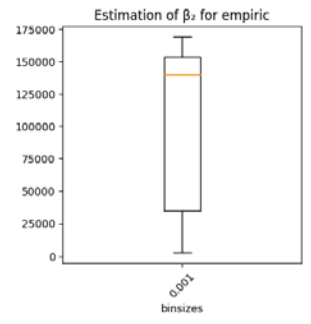


Figure 4.18

The figures above are box plots of whittles estimates made over 15 to 30 minute periods on real financial data. Each point corresponds to an estimate for a different day, the Hawkes process is estimated on transactions made around 11 a.m. . There are 21 estimates in all.

It seems that the parameters  $\alpha$  are estimated with the most precisions.

Since, we used MLE estimations as our initial conditions, we wanted to know if the estimations give us a lower value of the objective function to minimize. That's why we made a test of Student with the hypothesis than in average, the mean of the evaluation of the objective function to the new estimations are lower than the one for the MLE estimation. We observed a p-value of 0.48, so that the esimations by the whittle method are lower.

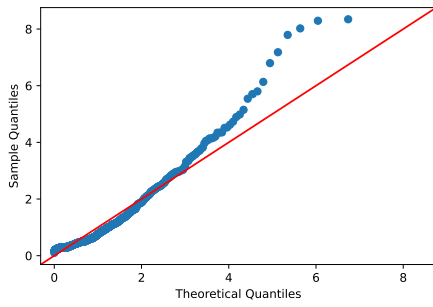


Figure 4.19: Fit: 2nd Jan, 11am-12pm

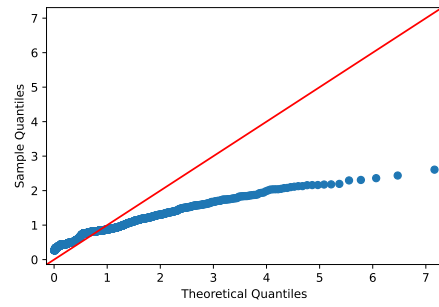


Figure 4.20: Fit: 2nd Jan, 11am-11:30pm

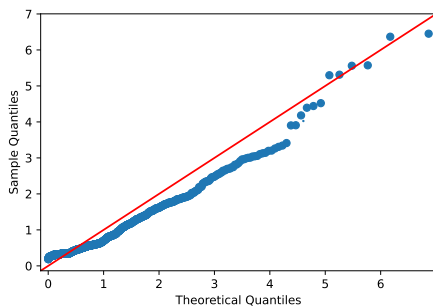


Figure 4.21: Fit: 3rd Jan, 11am-12pm

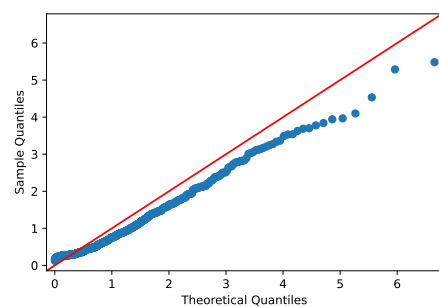


Figure 4.22: Fit: 2nd Jan, 11am-11:30pm

## 4.2.2 Unif Method

We employed the unif method to analyze the data across all hours of each day. To strike a balance between computational time and estimation precision, we opted for bin sizes of 0.1 and 0.01 for each hour. This compromise allowed us to achieve a reasonable level of precision while still considering the computational resources required. However, the results obtained using the unif method were not satisfactory. The plotted results represent the estimated data for the time interval between 11:00 am and 12:00 am on January 2nd. In fact, we observed similar outcomes to those obtained with the unif method using bin sizes of 0.1 and 0.01. Moreover, the computational time required

by the unif method rendered it impractical for practical use. Despite our efforts, there was no improvement in comparison to the MLE random method, making the unif method unsuitable for our analysis due to its lack of substantial progress and excessive computation time.

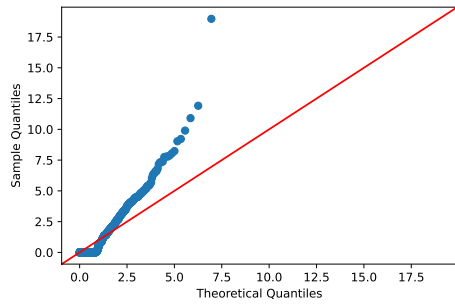


Figure 4.23: Fit: 2nd Jan, 11am-12pm (bin = 0.1)

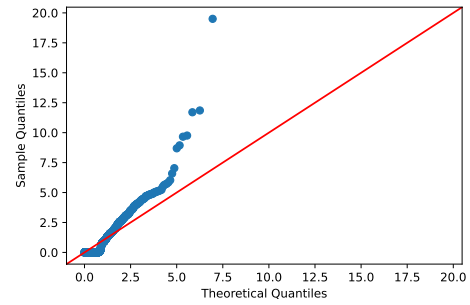


Figure 4.24: Fit: 2nd Jan, 11am-12pm (bin = 0.01)

## Chapter 5

# Conclusion

The project encompasses various aspects of estimating parameters of binned Hawkes processes, including one-exponential and two-exponential models.

In one-exponential Hawkes processes, the Whittle estimation method was utilized, along with the Uniform and Sequential methods. The Uniform method was employed first, followed by the Sequential method. Comparisons were made among these three methods, considering their respective advantages and limitations to reach the best tradeoff between accuracy and data precision constraints.

Moving on to two-exponential Hawkes processes, the Unif method was applied. This involved adapting the estimation principles to suit the two-exponential model, along with computing the Fourier transforms of the kernel and mean intensity. The Whittle estimation method was also adapted for two-exponential processes. The results derived from these methods demonstrated that, given an appropriate selection of the binsize, the parameter estimation could be remarkably accurate. Specifically, the Unif method exhibited superior performance compared to the Whittle method. However, as the binsize increased, the accuracy of the estimates significantly diminished. The Whittle method yielded biased estimates, while the Unif method proved inadequate in capturing the nuances of the two-exponential process.

Supervised learning techniques were explored, utilizing two specific model architectures: a dense traditional one and a LSTM network. The results demonstrated that the ranges of parameters and the bin size on which the models were trained had little influence on the performance, thus making these techniques very robust. We were able to show that NNs are capable of accurately estimating  $\mu$  and the branching ratio  $\eta$ , specifically for LSTM models. And even though we weren't capable to precisely estimate  $\beta$ , these methods seem very promising and we have to keep exploring new ways of training these models to have better performance.

The results and observations obtained from real data closely resembled those obtained from simulated data when fitting and modeling financial order using Hawkes processes. The **MLE\_random** method demonstrated a remarkably good fit, suggesting its applicability. However, the other methods did not exhibit the same level of effectiveness. In addition, it is important to note that the other methods employed in the study exhibited clear deficiencies. These shortcomings included for example long computation times, which contributed to their inferior performance compared to the **MLE\_random** method.

Overall, the project shed light on various estimation techniques such as the Whittle estimation, the MC-EM algorithm and supervised learning methods for binned Hawkes processes. The project's findings suggest that the **MLE\_random** method is the most effective approach for estimating binned Hawkes processes: this method was deemed accurate, quick, and free from constraints, thus outperforming all of its rival methods. Among these, only the Whittle estimation could be used in the case of degraded data to estimate  $\beta$  since **MLE\_random** becomes biased while Whittle remains centered but at the cost of a big variance.

# Appendix : Speeding up the optimization

In this section we try to speed up the call to the Python built-in minimization function of the *Scipy* library in Python used in the **Unif** method. The main motivation being our code is twice as slow as the same code executed on Matlab.

To do this, we first start by computing the gradient and the Hessian matrix of the cost function of the *unif* method in order to try to speed up the computations and avoid local minima, if they exist.

The cost function is

$$f(\mu, \alpha, \beta) = -\frac{1}{N_{MonteCarlo}} \sum_{i=1}^{N_{MonteCarlo}} \omega_i l_i(\mu, \alpha, \beta, T_i) \quad (1)$$

where  $l_i$  is the log-likelihood of the Hawkes process  $T_i$  on  $[0, T]$  and  $\omega_i$  the weight associated to the  $i$ th Monte-Carlo simulation. The expression of the log-likelihood is given by

$$l = \sum_{i=1}^k \ln(\lambda(t_i)) - \int_0^T \lambda(s) ds \quad (2)$$

where the  $t_i$  are the timestamps of the Hawkes process of conditional intensity  $\lambda$  given by

$$\lambda(t) = \mu + \sum_{t_i < t} \alpha \beta e^{-\beta(t-t_i)} \quad (3)$$

We know that (see our Whittle report)

$$l(\mu, \alpha, \beta) = \sum_{i=1}^k \ln(\mu + \alpha \beta \sum_{j=1}^{i-1} e^{-\beta(t_i-t_j)}) - \mu T + \alpha \sum_{i=1}^k (e^{-\beta(T-t_i)} - 1) \quad (4)$$

We consider the following quantity

$$A(i) = \sum_{j=1}^{i-1} e^{-\beta(t_i-t_j)} \text{ with } A(1) = 0 \quad (5)$$

Now, we compute the partial derivatives to find the gradient.

$$\frac{\partial l}{\partial \mu} = \sum_{i=1}^k \frac{1}{\mu + \alpha \beta A(i)} - T \quad (6)$$

$$\frac{\partial l}{\partial \alpha} = \sum_{i=1}^k \frac{\beta A(i)}{\mu + \alpha \beta A(i)} + \sum_{i=1}^k (e^{-\beta(T-t_i)} - 1) \quad (7)$$

$$\frac{\partial l}{\partial \beta} = \sum_{i=1}^k \frac{\alpha \sum_{j=1}^{i-1} e^{-\beta(t_i-t_j)} (1 - (t_i - t_j) \beta)}{\mu + \alpha \beta A(i)} - \alpha \sum_{i=1}^k (T - t_i) e^{-\beta(T-t_i)} \quad (8)$$

Lets rewrite the numerator of the first sum for a given  $i \in \{1, \dots, k\}$ :

$$\alpha \sum_{j=1}^{i-1} e^{-\beta(t_i - t_j)} (1 - (t_i - t_j)\beta) = \alpha(A(i) - \beta \sum_{j=1}^{i-1} (t_i - t_j) e^{-\beta(t_i - t_j)}) \quad (9)$$

We already know that (see Whittle report)

$$A(i) = e^{-\beta(t_i - t_{i-1})} (1 + A(i-1)) \text{ for } i \in \{1, \dots, k\} \quad (10)$$

This gives a first order induction formula which allows to compute A with linear complexity. In what follows we do the same to compute the last term of the sum above:

$$\begin{aligned} C(i) &= \sum_{j=1}^{i-1} (t_i - t_j) e^{-\beta(t_i - t_j)} \\ &= e^{-\beta t_i + \beta t_{i-1}} e^{\beta t_i - \beta t_{i-1}} \sum_{j=1}^{i-1} (t_i - t_j) e^{-\beta(t_i - t_j)} \\ &= e^{-\beta t_i + \beta t_{i-1}} \sum_{j=1}^{i-1} (t_i - t_j) e^{-\beta(t_{i-1} - t_j)} \\ &= e^{-\beta t_i + \beta t_{i-1}} \left( \sum_{j=1}^{i-2} (t_i - t_j) e^{-\beta(t_{i-1} - t_j)} + (t_i - t_{i-1}) \right) \\ &= e^{-\beta t_i + \beta t_{i-1}} (C(i-1) + (t_i - t_{i-1})) \end{aligned} \quad (11)$$

This shows that the sequence  $C$  can also be computed in linear cost.

We can rewrite

$$\frac{\partial l}{\partial \beta} = \sum_{i=1}^k \frac{\alpha A(i) - \alpha \beta C(i)}{\mu + \alpha \beta A(i)} - \alpha \sum_{i=1}^k (T - t_i) e^{-\beta(T - t_i)} \quad (12)$$

We obtain the following expression for the gradient of the cost function:

$$\nabla f(\mu, \alpha, \beta) = -\frac{1}{N_{MonteCarlo}} \sum_{i=1}^{N_{MonteCarlo}} \omega_i \nabla l_i(\mu, \alpha, \beta, T_i) \quad (13)$$

With the gradient computed, the idea was to apply either Newton-CG or L-BFGS-B algorithms for the minimization of the cost function. However, this did not work: the optimization was very dependent on the initial guess of parameters and converged to outlier parameters very often. So, we tried to implement a gradient descent "by hand" (meaning without any built-in Python function) but even then, the results were still not good (trying with both a constant learning rate and a dynamic learning rate).

Our next idea was to think that maybe by specifying the Hessian matrix, Newton-CG or L-BFGS-B would converge to the true parameters. Here are the computations needed for the Hessian matrix:

$$\frac{\partial^2 l}{\partial \mu^2} = -\sum_{i=1}^k \frac{1}{(\mu + \alpha \beta A(i))^2} \quad (14)$$

$$\frac{\partial^2 l}{\partial \mu \partial \alpha} = -\sum_{i=1}^k \frac{\beta A(i)}{(\mu + \alpha \beta A(i))^2} \quad (15)$$

$$\frac{\partial^2 l}{\partial \mu \partial \beta} = -\sum_{i=1}^k \frac{\alpha A(i)}{(\mu + \alpha \beta A(i))^2} \quad (16)$$

$$\frac{\partial^2 l}{\partial \alpha^2} = -\sum_{i=1}^k \frac{(\beta A(i))^2}{(\mu + \alpha \beta A(i))^2} \quad (17)$$

$$\frac{\partial^2 l}{\partial \alpha \partial \beta} = - \sum_{i=1}^k \frac{\mu A(i)}{(\mu + \alpha \beta A(i))^2} - \sum_{i=1}^k (T - t_i) e^{-\beta(T-t_i)} \quad (18)$$

$$\frac{\partial^2 l}{\partial \beta^2} = \sum_{i=1}^k \frac{\mu \alpha C(i) + (\alpha A(i))^2}{(\mu + \alpha \beta A(i))^2} + \alpha \sum_{i=1}^k (T - t_i)^2 e^{-\beta(T-t_i)} \quad (19)$$

By computing the Hessian matrix, our idea was to implement a Newton optimization algorithm. However, a necessary condition to apply this method is that the Hessian matrix is strongly positive. This is not the case since two of the three diagonal terms of the matrix are negative. So, this algorithm can't be applied.

To summarize, we weren't able to speed up the optimization step. The main motivation for this came from the fact that the Matlab code given in [6] performed *at least* twice as quick as in Python. But, the Matlab code used an interior-point algorithm which isn't implemented in any standard Python library. Only the Python library *gekko* performs minimizations with the interior point algorithm but it uses its own data structure which created many conflicts with the rest of our code. For this reason, we didn't manage to adapt our code to the data structure requirements of *gekko*. Thus, we didn't change our code and decided we would come back to this issue later on if it was really necessary.

# Bibliography

- [1] Yannick Bessy-Rol and Alice Launay. Modélisation du risque terroriste par les processus de hawkes. *EURIA, Euro-Institut d'Actuariat, mémoire de master*, 2017.
- [2] Gabriel Lang Felix Cheysson. Strong-mixing rates for hawkes processes and application to whittle estimation from count data. *arXiv:2003.04314*, 2021.
- [3] Tom Keane. Statistical inference for hawkes processes with deep learning. 2020.
- [4] Mehdi Lallouache and Damien Challet. The limits of statistical significance of hawkes processes fitted to financial data. *arXiv:1406.3967*, 2015.
- [5] Patrick Laub. Hawkes processes: Simulation, estimation, and validation. *University of Queensland*, 2014.
- [6] Niall Adams Lekha Patel Leigh Shlomovich, Edward A. K. Cohen. Parameter estimation of binned hawkes processes. *Journal of Computational and Graphical Statistics*, 2022.
- [7] Ioane Muni Toke and Fabrizio Pomponio. Modelling trades-through in a limit order book using hawkes processes. *Economics, Volume 6 Issue 1*, 2012.